

VRGaussianAvatar: Integrating 3D Gaussian Avatars into VR

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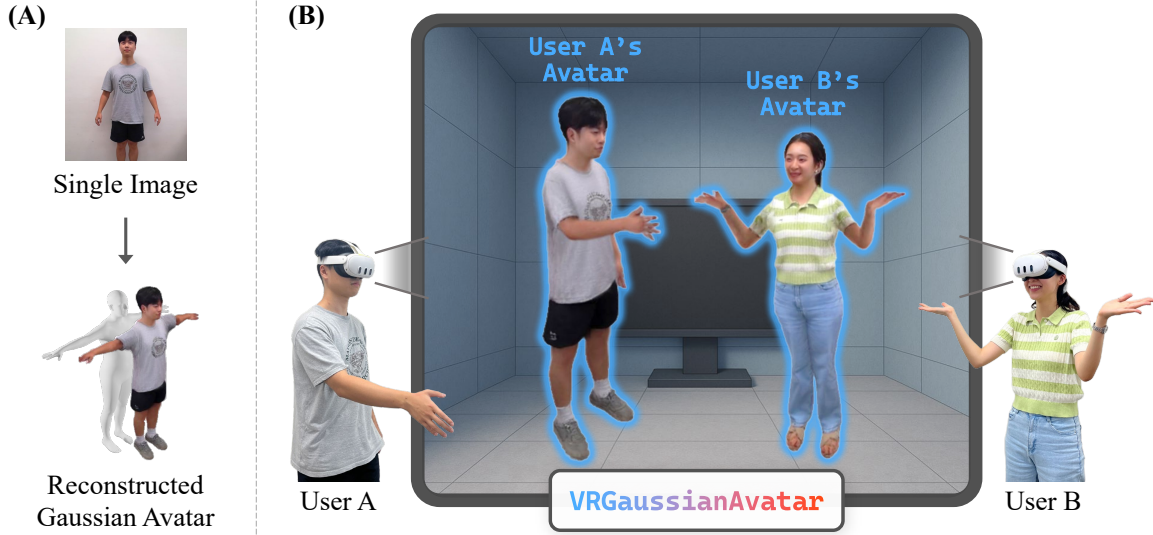


Fig. 1: We introduce **VRGaussianAvatar**, an integrated VR system for controllable full-body 3DGS avatar representation. (A) A single input image is used to reconstruct a full-body 3D Gaussian Splatting avatar. (B) The reconstructed avatar supports real-time rendering and full controllability in VR using only the internal sensors of commercial head-mounted displays (HMDs).

Abstract—We present VRGaussianAvatar, an integrated system that enables real-time full-body 3D Gaussian Splatting (3DGS) avatars in virtual reality using only head-mounted display (HMD) tracking signals. The system adopts a parallel pipeline with a VR Frontend and a GA Backend. The VR Frontend uses inverse kinematics to estimate full-body pose and streams the resulting pose along with stereo camera parameters to the backend. The GA Backend stereoscopically renders a 3DGS avatar reconstructed from a single image. To improve stereo rendering efficiency, we introduce Binocular Batching, which jointly processes left and right eye views in a single batched pass to reduce redundant computation and support high-resolution VR displays. We evaluate VRGaussianAvatar with quantitative performance tests and a within-subject user study against image- and video-based mesh avatar baselines. Results show that VRGaussianAvatar sustains interactive VR performance and yields higher perceived appearance similarity, embodiment, and plausibility. Project page and source code are available at <https://vrgaussianavatar.github.io>

Index Terms—Virtual reality, Augmented reality, 3D Gaussian Splatting, Avatar reconstruction, Stereoscopic rendering

1 INTRODUCTION

The representation of photorealistic full-body avatars in Virtual Reality (VR) plays a crucial role in enhancing user immersion and effective communication. Applications such as telepresence and gaming benefit significantly from avatars that faithfully replicate the user’s appearance.

To make avatar creation practical and accessible, existing methods

allow for the creation of mesh-based 3D avatars. These methods reconstruct drivable human models from a single image [3, 10, 20, 38, 59] or from video sequences [1, 23, 48]. While these approaches are highly compatible with existing graphics pipelines and offer strong real-time performance, they often struggle to deliver photorealistic avatar representations for subjects that significantly deviate from template meshes such as SMPL-X [38]. Recently, neural rendering techniques, including Neural Radiance Fields (NeRF) [34] and 3D Gaussian Splatting (3DGS) [21], have achieved remarkable success across diverse 3D visualization tasks. These methods have been applied not only to novel view synthesis but also to object reconstruction [24, 49, 50] and avatar creation [4, 12, 17]. In particular, 3DGS represents a 3D scene as a collection of colored Gaussian splats, enabling real-time photorealistic rendering and demonstrating strong performance in both face [13, 52, 65] and body avatar generation [35, 42, 45].

In VR, several studies [18, 53] have proposed techniques that leverage 3DGS for immersive scene representation. Kim et al. [22] conducted quantitative evaluations and user studies to examine the effectiveness of 3DGS-based object representation in VR, confirming that 3DGS is effective for realistic object rendering in virtual environments. VOODOO XP [51] was the first to propose a VR telepresence application using a 3DGS-based head avatar animated using the internal sensors of head-mounted displays (HMDs).

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VR高斯虚拟形象：将3D高斯虚拟形象整合到虚拟现实环境中

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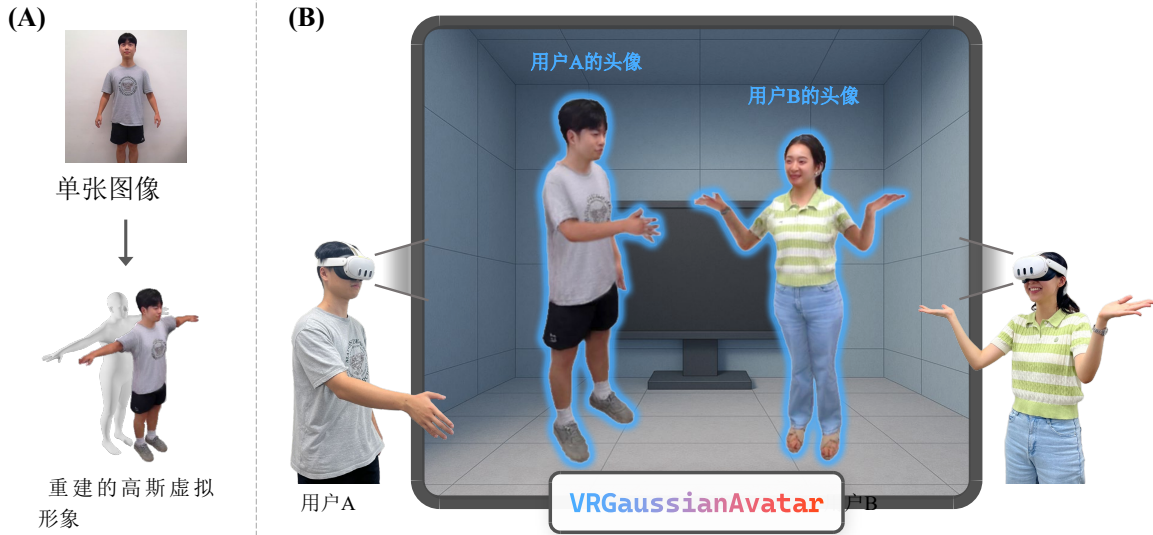


图1: 我们推出VR Gaussian Avatar, 这是一款用于实现可控全身3D Gaussian Splatting虚拟化身呈现的集成式VR系统。(A)仅需一张输入图像即可重建全身3D Gaussian Splatting虚拟化身。(B)该重建后的虚拟化身仅需用头戴式显示器(HMD)的内部传感器即可支持实时渲染及全维度操控功能。

摘要——我们提出VRGaussianAvatar, 这是一个集成系统, 仅需头戴式显示器(HMD)的跟踪信号即可在虚拟现实实现实时全身三维高斯贴图(3DGS)虚拟形象。该系统采用并行流水线架构, 包含VR前端和GA后端。VR前端利用逆运动学算法估算全身姿态, 并将所得姿态及立体相机参数传输至后端。GA后端通过立体视觉技术, 从单张图像重建出3DGS虚拟形象。为提升立体渲染效率, 我们引入了双眼批量处理技术, 通过单次批量处理左、右眼视图来减少冗余计算并支持高分辨率VR显示器。我们通过定量性能测试和受试者内用户研究, 将VRGaussianAvatar与基于图像和视频的网格虚拟形象基线进行对比。结果表明, VRGaussianAvatar能保持交互式VR性能, 并在视觉外观相似度、具身感和可信度方面表现更优。项目页面及源代码可访问<https://vrgaussianavatar.github.io>获取。

索引术语——虚拟现实、增强现实、3D高斯斑点技术、虚拟形象重建、立体渲染

1 引言

在虚拟现实(VR)中, 逼真度极高的全身虚拟形象的呈现对于提升用户的沉浸感和促进有效沟通至关重要。诸如远程呈现(telepresence)和游戏等应用, 都能从能够精确还原用户外貌的虚拟形象中获益良多。为了使虚拟形象的创建变得实用且易于操作, 现有的方法.....

支持创建基于网格的3D虚拟形象。这些方法能够从单张图像[3、10、20、38、59]或视频序列[1、23、48]重建可驱动的人体模型。尽管这些方法与现有图形处理流程高度兼容且具备出色的实时性能, 但对于显著偏离模板网格(如SMPL-X[38])的主体, 往往难以生成逼真的虚拟形象表征。近年来, 神经渲染技术——包括神经辐射场(NeRF)[34]和三维高斯喷溅(3DGS)[21]——在多种3D可视化任务中取得了显著成功。这些方法不仅应用于新型视图合成, 还广泛用于物体重建[24、49、50]及虚拟形象创建[4、12、17]。具体而言, 3DGS将三维场景表示为一组彩色高斯斑点集合, 能够实现实时照片级真实渲染, 并在面部[13、52、65]和身体虚拟形象生成[35、42、45]方面均展现出卓越性能。

在虚拟现实(VR)领域, 多项研究[18、53]提出了利用三维地理信息系统(3DGS)实现沉浸式场景呈现的技术。Kim等人.....[22]研究人员通过定量评估和用户研究, 检验了基于3DGS的对象表示在虚拟现实(VR)中的有效性, 证实3DGS能够实现虚拟环境中逼真的对象渲染效果。voodoo XP[51]是首个提出基于3DGS的VR远程呈现应用的系统, 该应用利用头戴式显示器(HMD)的内置传感器对头部虚拟形象进行动态动画处理。

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However, several challenges have prevented 3DGS-based full-body avatars from being deployed in VR communication systems. In addition to the significant hardware demands of 3DGS visualization, VR requires stereoscopic rendering which imposes even higher computational demands. Moreover, 3DGS pipelines have limited compatibility with conventional VR rendering pipelines. Real-time animation of 3DGS avatars remains challenging, and existing systems do not directly utilize user motion captured from the internal sensors of HMDs. These limitations have hindered the integration of 3DGS avatars into VR.

To overcome these limitations, we present **VRGaussianAvatar**, an integrated system that enables real-time user representation in VR through a 3DGS avatar reconstructed from a single image and controllable using only the internal sensors of a commercial HMD. Our system consists of two subsystems: a VR Frontend, which processes sensor data from a commercial HMD and applies inverse kinematics for avatar animation, and a GA Backend (Gaussian Avatar Backend), which performs real-time photorealistic rendering with 3DGS. To support stereoscopic VR rendering, we introduce **Binocular Batching**, which achieves efficient and consistent rendering for both eyes. The system builds on standard components (e.g., off-the-shelf inverse kinematics and recent single-image 3DGS avatar reconstruction models), while our main contributions are the VR streaming integration, Binocular Batching, and a perceptual user study of 3DGS avatars in VR.

We validate our system through user studies and quantitative evaluations comparing VRGaussianAvatar with mesh-based avatars reconstructed from images and videos. These mesh avatars were built upon the parametric whole-body model SMPL-X [38] and were designed to be controlled solely through the internal sensors of the HMD, similar to ours. For the user study, subjects followed instructions to control both baseline and our method’s avatars, modeled after themselves and others. They then completed questionnaires assessing virtual embodiment and plausibility. The results show that our method provides stronger self-embodiment than mesh-based baselines. Quantitative evaluations further confirm that avatars produced by our approach resemble the user’s real appearance more closely.

In summary, our contributions are as follows:

- **VRGaussianAvatar**: an integrated VR streaming system for stereoscopic rendering of a 3DGS full-body avatar driven by HMD-only tracking.
- **Binocular Batching**: an efficient stereoscopic rendering strategy that jointly renders the left and right eye views in a single batched pass for 3DGS avatars in VR.
- A quantitative evaluation and perceptual user study showing improved perceived appearance similarity, embodiment, and plausibility over mesh-based avatar baselines under the same HMD-only control setting.

2 RELATED WORK

2.1 Avatar Mediated Communication

In VR environments, virtual embodiment and avatar plausibility during communication are both crucial. For effective VR communication, it is important not only that individuals identify with their own avatars through virtual embodiment, but also that avatars are perceived as plausible in appearance and behavior. Previous studies have shown that visual appearance plays a key role in virtual embodiment and more realistic appearances resembling actual humans significantly enhance embodiment [56]. To achieve high levels of plausibility in VR communication, avatar realism is crucial. Studies have concluded that high-quality avatars enhance plausibility-related judgments and user comfort. For example, Yoon et al. [60] demonstrated that realistic full-body avatars are advantageous regarding virtual embodiment and plausibility. Similarly, Cho et al. [6] showed that realistic volumetric avatars yield higher perceived plausibility than pre-scanned avatars when performing both dynamic and static tasks. Moreover, Kaiser et al. [19] reported through quantitative and qualitative experiments that higher avatar realism generally improves plausibility and comfort.

In this work, we present a VR avatar representation system that leverages a full-body avatar based on 3D Gaussian Splatting (3DGS), which

has recently shown strong performance in realistic 3D human body model representation. Furthermore, through user studies and statistical analysis, we confirm that the proposed Gaussian avatar representation system contributes to increasing users’ virtual embodiment and avatar plausibility.

2.2 3D Avatar Reconstruction

The generation of 3D avatars resembling users has been a critical topic in the areas of computer vision, computer graphics, and virtual reality. For full-body avatar reconstruction, the parametric body model SMPL-X [38] has been widely adopted, as it encodes body shape and facial expressions into latent features and enables pose-dependent 3D representations and animations driven by body, hand, and facial configurations. Using SMPL-X as a template mesh, several studies have explored approaches to reconstruct mesh and texture maps from videos [1, 48, 67] or images [28, 38].

The advancement of neural rendering techniques, represented by neural radiance fields (NeRF) [34], has demonstrated strong performance in generating photorealistic full-body avatars from videos [4, 17, 61, 66] or images [14]. However, the high computational cost and long processing times of these methods limit their applicability to real-time avatar representation in VR environments. To achieve realistic 3D representations with efficient rendering, 3DGS was introduced as a neural rendering technique designed for both fidelity and speed. 3DGS enables fast and photorealistic avatar representation from videos [27, 35, 36, 39, 41] or image [32, 42, 43, 45] and has also shown superior performance in head model reconstruction [5, 40, 46, 51, 52, 64, 65] compared to NeRF-based approaches.

In particular, the Large Human Model (LHM) [42] achieves one-shot avatar reconstruction in just a few seconds while supporting real-time rendering on consumer-grade GPUs. We adopt LHM [42] as an off-the-shelf backbone to reconstruct 3D Gaussian avatars, and focus on integrating it into a practical VR streaming and stereoscopic rendering system.

2.3 VR Avatar Representation Using Neural Rendering

There have been several studies that explored the use of neural rendering for photorealistic scene and object representation in VR. Immersive-NGP [25], which applied the hash-table-based instant-NGP [37] to accelerate NeRF rendering to near real-time, was the first work to demonstrate an application of neural rendering in VR. VR-GS [18] further leveraged 3DGS, which enables both faster rendering and more realistic 3D representations, to achieve real-time VR rendering and introduced physics-based interactions. However, these works have been limited in that they cannot be directly applied to neural rendering-based VR avatar representation, which requires real-time rendering of dynamically moving neural primitives.

To address this, SqueezeMe [16] reduced the computational overhead of animation and 3DGS rendering, making real-time full-body avatar representation in VR. For head avatars, VOODOO XP [51] demonstrated the animation and rendering of a 3DGS head avatar driven by the internal sensors of HMD. Nevertheless, no prior work has realized a full-body avatar representation using only the limited information from HMD-based body tracking and head tracking. Motivated by these gaps, we integrate a full-body 3DGS avatar into a VR streaming pipeline and evaluate its perceptual impact in VR.

3 METHOD

VRGaussianAvatar is structured as a parallel framework with a **VR Frontend** and a **GA Backend**. The VR Frontend estimates full-body pose from HMD sensors and provides per-frame camera parameters. The GA Backend animates and renders the pre-reconstructed 3DGS avatar, incorporating a **Binocular Batching** strategy for efficient stereoscopic rendering. In our reference setup, the VR Frontend and GA Backend communicate over localhost to minimize network overhead. Figure 2 presents the overall system architecture.

然而,若干技术挑战阻碍了基于3DGS技术的全身虚拟形象在VR通信系统中的广泛应用。除了3DGS可视化技术对硬件资源的高要求外,VR还需要实现立体渲染,这进一步增加了计算负担。此外,3DGS技术流程与传统VR渲染流程的兼容性有限。3DGS虚拟形象的实时动画效果仍存在技术瓶颈,现有系统也无法直接利用头戴式显示器(HMD)内置传感器捕捉的用户动作数据。这些局限性严重制约了3DGS虚拟形象在VR领域的集成应用。

为克服这些局限性,我们推出了**VR Gaussian Avatar**——一个集成系统,能够通过仅使用商用头戴式显示器(HMD)的内部传感器,从单张图像重建3DGS虚拟形象,并实现实时的虚拟现实用户呈现。该系统由两个子系统组成:VR前端负责处理商用HMD的传感器数据,并运用逆运动学算法实现虚拟形象动画;GA后端(Gaussian Avatar Backend)则利用3DGS技术进行实时高逼真渲染。为支持立体VR渲染,我们引入了**双眼批量渲染技术**,可实现双眼高效且一致的渲染效果。该系统基于标准组件(如现成的逆运动学算法及最新的单图像3DGS虚拟形象重建模型),其主要创新点包括VR流媒体集成技术、双眼批量渲染机制,以及针对VR中3DGS虚拟形象的感知用户体验研究。

我们通过用户研究和定量评估来验证系统性能,将VR Gaussian Avatar与基于图像和视频重建的网格虚拟形象进行对比。这些网格虚拟形象基于参数化全身模型 SMPL-X[38]构建,设计上仅能通过头戴式显示器(HMD)的内部传感器进行操控,与我们的系统类似。在用户研究中,受试者按照指示操控了基于自身及他人形象构建的基线虚拟形象和本方法生成的虚拟形象,随后填写了评估虚拟具身感与真实度的问卷。结果表明,我们的方法比基于网格的基线方法能提供更强的自我具身体验。定量评估进一步证实,本方法生成的虚拟形象与用户的实际外貌更为接近。总之,我们的贡献如下:

- **VRGaussianAvatar**: 一款集成式VR流媒体系统,专为基于头戴式显示器(HMD)追踪技术实现3DGS全身虚拟形象的立体渲染而设计。
- **双目批量渲染**: 一种高效的立体渲染策略,可为虚拟现实(VR)中的3DGS虚拟形象在单次渲染过程中同时生成左眼和右眼视图。
- 一项定量评估与感知用户研究表明,在相同的仅使用头戴式显示器(HMD)的控制环境下,相较于基于网格的虚拟形象基准模型,该虚拟形象在感知外观相似性、具身感及可信度方面均有所提升。

2 相关工作

2.1 通过阿凡达实现的沟通

在虚拟现实环境中,虚拟具身感与沟通时虚拟形象的可信度都至关重要。要实现有效的VR沟通,不仅需要用户通过虚拟具身感与自身虚拟形象产生认同感,更要求这些虚拟形象在外观和行为上显得真实可信。先前研究表明,视觉外观对虚拟具身感起着关键作用——更接近真实人类的逼真外观能显著增强具身感[56]。要实现高水平的VR沟通可信度,虚拟形象的真实感尤为关键。研究证实,高质量的虚拟形象能提升用户对可信度的判断并增强使用舒适度。例如,Yoon等人.....[60]研究表明,逼真的全身虚拟化身在虚拟具身感和可信度方面具有显著优势。类似地,Cho等人也得出了相同结论。[6]研究表明,在执行动态和静态任务时,逼真的体积虚拟形象比预先扫描的虚拟形象能产生更高的感知可信度。此外,Kaiser等人.....[19]定量与定性实验表明,更高的虚拟形象真实度通常能提升其可信度与使用舒适度。

在本研究中,我们提出了一种基于三维高斯点扩散技术(3DGS)的全身虚拟化身表示系统。

该系统近期在逼真的人体3D模型呈现方面展现出优异性能。此外,通过用户研究和统计分析,我们证实所提出的高斯虚拟形象表示系统能够有效提升用户的虚拟身体感知体验及虚拟形象的真实感。

2.2 3D虚拟形象重建

生成与用户高度相似的3D虚拟形象,一直是计算机视觉、计算机图形学和虚拟现实领域的重要研究课题。在全身虚拟形象重建方面,参数化身体模型 SMPL-X[38]已被广泛采用,因为它能将身体形态和面部表情编码为潜在特征,并支持基于身体、手部及面部构型生成姿态相关的3D模型与动画。以 SMPL-X作为模板网格,多项研究探索了从视频[1, 48, 67]或图像[28, 38]重建网格与纹理贴图的方法。

以神经辐射场(NeRF)[34]为代表的神经渲染技术,在从视频[4, 17, 61, 66]或图像[14]生成逼真全身虚拟形象方面展现出卓越性能。然而,这些方法高昂的计算成本和漫长的处理时间限制了其在VR环境中实时虚拟形象呈现的适用性。为实现高效渲染下的真实3D表现,3DGS作为一种兼具高保真度与高速度的神经渲染技术应运而生。3DGS能够快速生成基于视频[27, 35, 36, 39, 41]或图像[32, 42, 43, 45]的逼真虚拟形象,并且在头部模型重建方面[5, 40, 46, 51, 52, 64, 65]相较于基于NeRF的方法表现出更优性能。

特别是,大型人体模型(LHM)[42]仅需数秒即可完成单次虚拟形象重建,并支持在消费级GPU上进行实时渲染。我们采用LHM[42]作为现成的骨干网络来重建三维高斯虚拟形象,并重点将其集成到实用的VR流媒体与立体渲染系统中。

2.3 基于神经渲染的VR虚拟化身表示方法

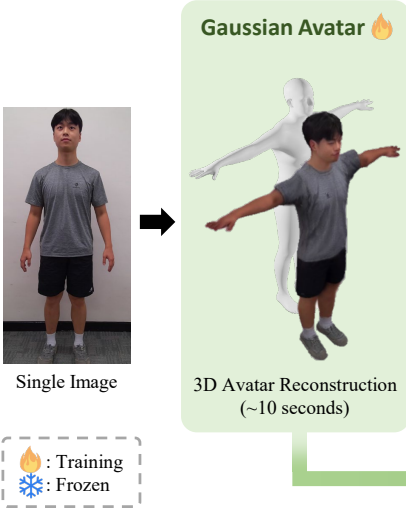
已有多项研究探讨了神经渲染技术在虚拟现实(VR)中实现逼真场景与物体表现的应用。ImmersiveNGP[25]率先将基于哈希表的即时 NGP[37]技术应用于NeRF渲染,使其达到近乎实时的处理速度,成为首个展示神经渲染在VR领域实际应用的研究成果。VR-GS[18]则进一步利用3DGS技术——该技术既能提升渲染速度,又能生成更逼真的三维模型——实现了实时VR渲染并引入了基于物理的交互功能。然而,这些研究存在局限性:它们无法直接应用于基于神经渲染的VR虚拟形象表现,因为后者需要对动态运动的神经图元进行实时渲染。

为解决这一问题,SqueezeMe[16]降低了动画与3DGS渲染的计算开销,实现了虚拟现实中的实时全身虚拟形象呈现。对于头部虚拟形象,voodoo XP[51]展示了基于头戴式显示器(HMD)内部传感器驱动的3DGS头部虚拟形象的动画与渲染技术。然而,此前尚无研究仅利用基于HMD的身体追踪和头部追踪的有限信息实现全身虚拟形象呈现。针对这些技术空白,我们将全身3DGS虚拟形象整合到VR流媒体处理流程中,并评估了其在虚拟现实中的感知效果。

3 方法

VRGaussianAvatar采用并行架构,包含**VR前端**和**GA后端**。VR前端通过头戴式显示器(HMD)传感器估算全身姿态,并提供逐帧相机参数。GA后端对预重建的3DGS虚拟形象进行动画渲染,采用**双目批量处理策略**以实现高效的立体渲染。在我们的参考配置中,VR前端与GA后端通过localhost进行通信,以最小化网络开销。图2展示了系统的整体架构。

Avatar Reconstruction (Offline)



Runtime Process (Online)

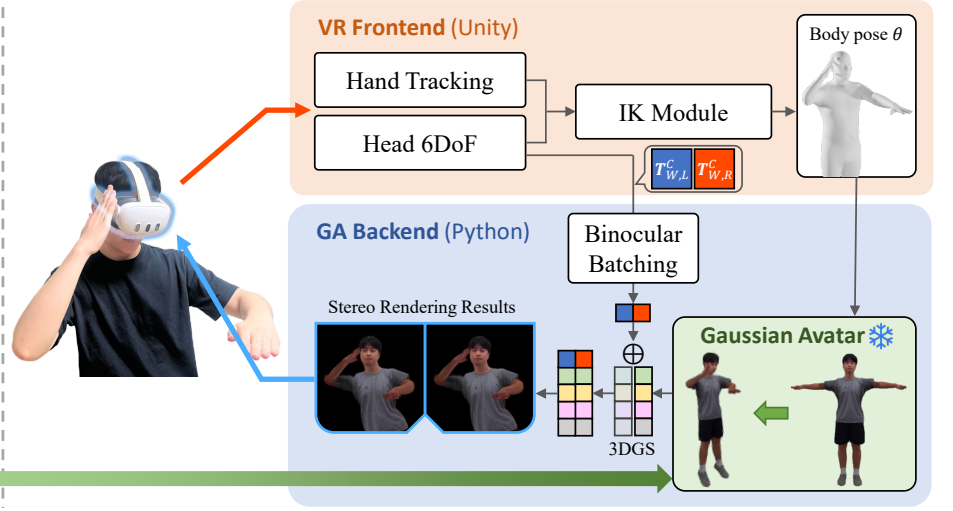


Fig. 2: System diagram of the proposed method, **VRGaussianAvatar**. *Avatar reconstruction phase (left)*: Given a single input image, the Gaussian Avatar Module reconstructs a 3D full-body avatar. *Runtime process (right)*: The **VR Frontend** and **GA Backend** operate in parallel to animate and render the avatar in real time. The VR Frontend estimates a SMPL-X-compatible full-body pose from head 6DoF and hand-tracking signals using the IK Module. The GA Backend animates and renders the reconstructed Gaussian avatar, and applies the proposed **Binocular Batching** with both world-to-camera matrices ($\mathbf{T}_{W,L}^C, \mathbf{T}_{W,R}^C$) to enable efficient real-time stereoscopic rendering.

3.1 VR Frontend

Inputs and Objective. The VR Frontend uses only device-native tracking signals from a commercial HMD: head and hand poses (position and orientation). Its goal is to estimate a full-body pose at interactive rates from these sparse cues and to stream per-frame stereo camera parameters to the GA Backend.

Inverse Kinematics Module (IK Module). The VR Frontend reconstructs a full-body pose using an off-the-shelf IK solver. The head pose provides the root constraint via a calibrated head-to-root offset. When available, left/right hand tracking is used as end-effector constraints to stabilize upper-limb articulation; otherwise, the solver falls back to a head-only configuration with posture priors on the torso and shoulders. The solver enforces joint limits and applies temporal smoothing for stability. The output is a full-body pose parameter set θ (SMPL-X compatible) and the head pose $\mathbf{T}_H^W$ used to derive stereo cameras.

Stereo Camera Construction. From the head pose, we construct per-eye camera transforms and transmit them to the GA Backend. We place an interpupillary baseline IPD along the head local +X axis:

$$\mathbf{T}_L^H = \begin{bmatrix} \mathbf{I} & (-\text{IPD}/2, 0, 0)^\top \\ \mathbf{0}^\top & 1 \end{bmatrix}, \quad \mathbf{T}_R^H = \begin{bmatrix} \mathbf{I} & (\text{IPD}/2, 0, 0)^\top \\ \mathbf{0}^\top & 1 \end{bmatrix}. \quad (1)$$

World-space eye poses are obtained by composition:

$$\mathbf{T}_e^W = \mathbf{T}_H^W \mathbf{T}_e^H, \quad e \in \{L, R\}. \quad (2)$$

The GA Backend uses the corresponding world-to-camera transforms:

$$\mathbf{T}_{W,e}^C = (\mathbf{T}_e^W)^{-1}. \quad (3)$$

We use the standard pinhole camera model to derive per-eye intrinsics and projection from the headset-provided field of view and render resolution.

Per-Frame Packaging. For each frame, the VR Frontend sends a compact request $\{\text{frame_id}, (\mathbf{T}_{W,e}^C, \mathbf{K}), \theta\}$ to the GA Backend, where frame_id is monotonically increasing, $e \in \{L, R\}$, $\mathbf{K}$ denotes shared camera intrinsics, and θ is the estimated body pose.

Implementation Details. In our reference implementation, the VR Frontend is developed with Unity 6000.1.13f1¹. The IK Module employs Unity Final IK². We use the Meta Quest 3³ as the HMD.

3.2 GA Backend

Role and Interface. The GA Backend receives per-frame requests from the VR Frontend and renders a 3DGS avatar reconstructed from a single image. Each request includes a sequential `frame_id`, per-eye camera parameters for L and R (world-to-camera transforms $\mathbf{T}_{W,e}^C \in \text{SE}(3)$ for $e \in \{L, R\}$ and a shared intrinsic matrix $\mathbf{K}$), and a full-body pose estimate. The backend returns a stereo pair that is streamed directly to the HMD.

Binocular Batching. We introduce Binocular Batching, which processes the two eye views of a frame within a single batch over a shared 3DGS state. In contrast to a naive stereo pipeline that renders the left and right eye sequentially with duplicated computation, our method stages per-Gaussian attributes $(p_i, \sigma_i, r_i, p_i, f_i)$ only once in GPU memory and reuses them across both eyes. Eye-specific projection and visibility $(\mathbf{K}, \mathbf{T}_{W,e}^C)$ are then evaluated separately within the same pass. This reduces redundant memory transfers and per-frame fixed overhead (e.g., kernel launches, data marshalling, cache warm-up), while improving cache coherence and memory locality. As a result, both eye buffers are produced in a single batched pass, yielding a notable reduction in rendering time, which we analyze in detail in subsection 5.2.

Gaussian Avatar Module. The Gaussian Avatar Module estimates canonical 3D Gaussian parameters, SMPL-X rig weights, and animation-ready bindings from a single image. In our implementation, this module is instantiated using the Large Animatable Human Model (LHM) [42] as an off-the-shelf reconstructor. The rest of the GA Backend is agnostic to the reconstructor, and any method that outputs $(\chi, \{w_{ik}\}, \{T_k(\cdot)\})$ can be plugged in.

¹<https://unity.com>

²<https://assetstore.unity.com/packages/tools/animation/final-ik-14290>

³<https://store.facebook.com/quest/products/quest-3>

化身重建（离线）

实时处理（在线）



图2：所提出方法VRGaussianAvatar的系统示意图。Avatar重建阶段（左）：给定单张输入图像，高斯Avatar Module用于重建三维全身虚拟形象。运行流程（右侧）：VR前端与GA后端并行运行，实时完成虚拟形象的动画渲染。VR前端通过IK Module根据头部6DoF数据和手部追踪信号估算出 SMPL-X兼容的全身姿态。GA后端对重建后的高斯虚拟形象进行动画渲染，并应用提出的双眼批量处理技术（结合世界坐标系至相机坐标系矩阵 TCW_L , TCW_R ），以实现高效的实时立体渲染。

3.1 VR Frontend

输入与目标。VR前端仅使用商用头戴式显示器（HMD）提供的设备原生跟踪信号：头部与手部的姿态（位置与方向）。其目标是基于这些稀疏线索以交互速率估算全身姿态，并将每帧立体相机参数流式传输至遗传算法后端。

逆运动学模块（IK Module）。VR前端通过商用逆运动学求解器重建全身姿态。头部姿态通过校准的头部-根部偏移量提供根约束；当条件允许时，利用左右手追踪数据作为末端执行器约束以稳定上肢关节运动；否则求解器将退而求其次，仅采用头部姿态参数，并为躯干和肩部设置姿态先验。该求解器会强制执行关节运动限制并应用时间平滑技术以提升稳定性。输出结果包括全身体态参数集 θ （兼容 SMPL-X格式）以及用于推导立体相机的头部姿态 TWH 。

立体相机构建。根据头部姿态，我们构建每只眼睛的相机变换参数，并将其传输至遗传算法后端。我们在头部局部坐标系的 X 轴方向上放置了一条瞳间距基线 IPD：

$$T_{HL} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} T(-IPD/21, 0, 0)^T, \quad T_{HR} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} T(IPD/21, 0, 0)^T. \quad (1)$$

世界空间眼位姿态是通过合成获得的：

$$T_{We} = T_{WH} T_{He}, \quad e \in \{L, R\}. \quad (2)$$

GA后端使用相应的世界到相机变换：

$$T_{W,e}^C = (T_e^W)^{-1}. \quad (3)$$

我们采用标准针孔相机模型，根据头戴设备提供的视场和渲染分辨率，推导出每只眼睛的内参参数及投影信息。

帧级封装。对于每一帧，VR前端会向GA后端发送一个紧凑的请求 $\{\text{frame_id}, (TCW, e, K), \theta\}$ ，其中 frame_id 呈单调递增， $e \in \{L, R\}$ ， K 表示共享相机内参， θ 为估计的身体姿态。

实现细节。在我们的参考实现中，VR前端采用Unity 6000.1.13f1¹开发；IK模块使用Unity Final IK²；我们采用Meta Quest 3³作为头戴式显示器（HMD）。

3.2 GA 后端

角色与接口。GA后端接收来自VR前端的逐帧请求，并基于单张图像重建3DGS虚拟形象进行渲染。每个请求包含顺序帧ID、 L 与 R 两眼的相机参数（世界坐标系到相机坐标系的变换 TCW ， $e \in \{L, R\}$ （适用于 $e \in \{L, R\}$ ），以及共享的本征矩阵 K ），以及全身姿态估计值。后端返回的立体图像会对直接流式传输至头戴式显示器（HMD）。

双目批量处理。我们提出双目批量处理技术，该技术在共享的3DGS状态下，将单帧中的双眼视图统一处理为一个批次。与传统的立体渲染管线（需顺序渲染左右眼并重复计算）不同，我们的方法仅在GPU内存中一次性处理每个高斯属性（ $\pi, \sigma_i, r_i, \rho_i, f_i$ ），并在两个眼睛间复用这些属性。随后在同一处理流程中分别计算各眼特有的投影与可见性参数（ K, TCW, e ）。这种方法减少了冗余内存传输和每帧固定开销（如内核调用、数据打包、缓存预热），同时提升了缓存一致性与内存局部性。最终，两个眼睛的缓冲区可在单次批量处理中生成，显著缩短渲染时间——我们将在5.2小节中对此进行详细分析。

高斯头像模块。 高斯化身模块

该模块可从单张图像中提取标准的3D高斯参数、SMPL-X骨骼权重以及适用于动画的绑定数据。在我们的实现中，此模块采用现成的大型可动画人体模型（LHM）[42]作为重建器进行实例化。GA后端其余部分对重建器无特定要求，任何能输出 $(\chi, \{wik\}, \{Tk(\cdot)\})$ 的方法均可直接集成使用。

¹<https://unity.com>

²<https://assetstore.unity.com/packages/tools/animation/final-ik-14290>

³<https://store.facebook.com/quest/products/quest-3>

Tokenization : We encode SMPL-X geometry and image appearance into transformer tokens. Let $X = \{x_j\}_{j=1}^{N_{pts}} \subset \mathbb{R}^3$ be canonical SMPL-X surface samples. Here, N_{pts} denotes the number of sampled surface points, which we set to 2,048 in our implementation. After a sinusoidal positional encoding $\gamma: \mathbb{R}^3 \rightarrow \mathbb{R}^{3L}$ and a modality-specific projection to the channel dimension C , we obtain

$$T^{3D} = \text{MLP}_{\text{proj}}(\gamma(X)) \in \mathbb{R}^{N_{pts} \times C}. \quad (4)$$

From the input image $I \in \mathbb{R}^{H \times W \times 3}$, a frozen body encoder E_{body} produces features in $\mathbb{R}^{N_{\text{body}} \times C_b}$ which are projected to

$$T^{\text{body}} = \text{MLP}_{\text{proj}}(E_{\text{body}}(I)) \in \mathbb{R}^{N_{\text{body}} \times C}. \quad (5)$$

To retain fine facial details, a frozen multi-scale head encoder yields $\{E_{\text{head}}^{(d)}(I) \in \mathbb{R}^{N_{\text{head}}^{(d)} \times C_{h,d}}\}_{d=1}^D$; a fusion module aggregates them into

$$T^{\text{head}} = F_{\text{fusion}}(\{E_{\text{head}}^{(d)}(I)\}_{d=1}^D) \in \mathbb{R}^{N_{\text{head}} \times C}, \quad (6)$$

where N_{head} denotes the post-fusion token count.

Prediction : A Multimodal Body-Head Transformer (MBHT) fuses the three token sets and a decoder predicts canonical Gaussians and rig parameters:

$$(\chi, \{w_{ik}\}, \{T_k(\cdot)\}) = \text{Decoder}(\text{MBHT}(T^{3D}, T^{\text{body}}, T^{\text{head}})). \quad (7)$$

Here, χ contains N Gaussians, $w_{ik} \geq 0$ with $\sum_k w_{ik} = 1$, and $T_k(\theta) \in \text{SE}(3)$ are rigid bone transforms defined in the canonical space.

Concretely, the canonical Gaussian set is

$$\chi = \left\{ (p_i, \sigma_i, r_i, \rho_i, f_i) \right\}_{i=1}^N, \quad (8)$$

where $p_i \in \mathbb{R}^3$ (centroid), $\sigma_i \in \mathbb{R}^3$ (anisotropic scales), $r_i \in \mathbb{S}^3$ (unit quaternion), $\rho_i \in [0, 1]$ (opacity), and $f_i \in \mathbb{R}^C$ (SH appearance). The 3D covariance is

$$\Sigma_i = R(r_i) \text{diag}(\sigma_i^2) R(r_i)^\top, \quad R(r_i) \in \text{SO}(3), \quad (9)$$

with $R(\cdot)$ the quaternion-to-rotation map. To enable animation, the module provides an SMPL-X-compatible rig with blend weights w_{ik} and bone transforms $T_k(\theta) \in \text{SE}(3)$ for pose θ .

Pose conditioning (runtime) : Given the per-frame pose parameters θ from the VR Frontend, we deform the canonical Gaussian set χ into a posed set $\chi' = \text{LBS}(\chi, \theta)$ using linear blend skinning (LBS) with per-Gaussian weights w_{ik} . We define the LBS output for Gaussian i as $(p'_i, \Sigma'_i) = \text{LBS}(p_i, \Sigma_i; \theta)$. Specifically, letting $T_k(\theta) = (R_k(\theta), t_k(\theta))$ denote the pose-dependent transform of bone k and $R_i(\theta)$ the induced blended rotation, we have

$$p'_i = \sum_k w_{ik} (R_k(\theta) p_i + t_k(\theta)), \quad (10)$$

$$\Sigma'_i = R_i(\theta) \Sigma_i R_i(\theta)^\top. \quad (11)$$

Splatting and compositing : We render posed Gaussians using the standard 3D Gaussian Splatting rasterization [21].

$$\mathbf{I}_e = \text{Render}(\text{LBS}(\chi, \theta), \mathbf{K}, \mathbf{T}_{W,e}^C), \quad e \in \{L, R\}, \quad (12)$$

where $\text{LBS}(\chi, \theta)$ denotes the posed Gaussian set after the LBS step above, and $(\mathbf{K}, \mathbf{T}_{W,e}^C)$ are the per-eye camera parameters. Under Binocular Batching, per-Gaussian attributes are staged once and reused for both L and R within a single batch, while per-eye projection and view-dependent shading are evaluated per eye in the same pass.

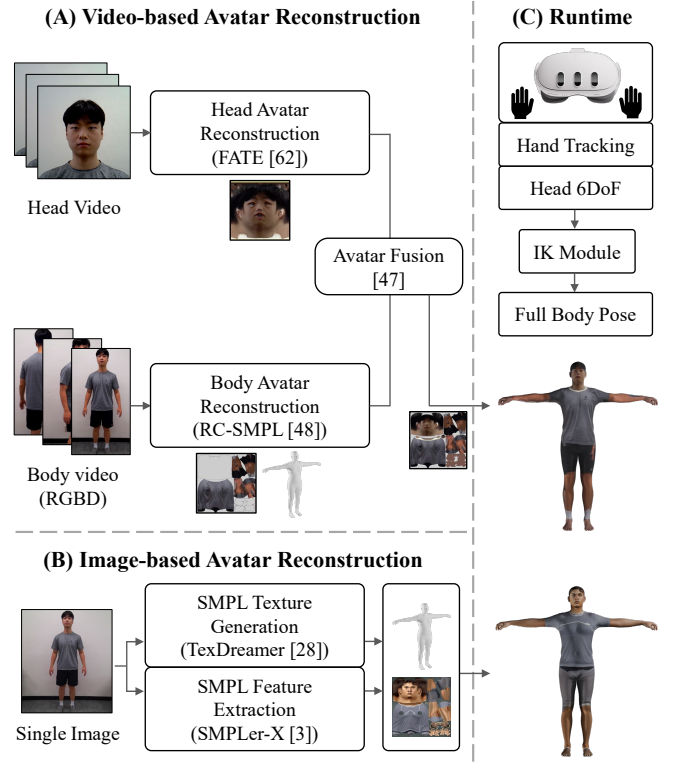


Fig. 3: **Baseline conditions.** (A) *video-based reconstruction*: Implemented following prior works [47, 48, 62]. (B) *image-based reconstruction*: Implemented following prior works [3, 28]. (C) *Runtime process*: The IK Module estimates a full-body pose from head 6DoF and hand-tracking signals to animate the avatar in real time.

Encoding and Delivery to HMD. Rendered frames are converted to 8-bit sRGB and JPEG-compressed at a fixed quality; the left/right payloads, together with `frame_id`, are transmitted in a single response and delivered directly to the HMD. The stateless per-frame design admits deterministic dropping of late frames on the receiver.

The GA Backend renders the 3DGS avatar with an alpha channel to support optional avatar compositing when a 3DGS scene/background is included. In our study, we disable background compositing and use a black background to minimize background influence and focus the evaluation on the avatar itself; accordingly, we stream RGB-only JPEG frames to reduce bandwidth.

Algorithmic Outline.

1. Receive $\{\text{frame_id}, (\mathbf{T}_{W,e}^C, \mathbf{K}), \theta\}$ from the VR Frontend.
2. Construct a single-batch stereo rendering context and apply pose conditioning to the canonical 3DGS avatar, deforming Gaussian primitives according to θ .
3. Execute one batched splatting pass with *Binocular Batching* to simultaneously render left and right eye images $[L, R]$.
4. JPEG-compress the two rendered eye buffers and package them with `frame_id`.
5. Stream the compressed response directly back to the HMD for display.

Implementation Details. GA Backend is implemented in Python 3.9⁴, PyTorch 2.3.0⁵, and PyTorch3D⁶. Rendering runs on a single NVIDIA RTX 4090 GPU in real time.

⁴<https://www.python.org/>

⁵<https://pytorch.org/>

⁶<https://pytorch3d.org/>

分词：我们将 SMPL-X 的几何结构和图像外观编码为 Transformer 分词符。设 $X = \{x_j | N_{pts} = 1 \dots R^3$ 中的样本为标准的 SMPL-X 曲面样本。此处， N_{pts} 表示采样的曲面点数量，在我们的实现中将其设为 2,048。经过正弦位置编码 $\hat{d}: R^3 \rightarrow R^{3 \times L}$ 以及针对通道维度 C 的模态特定投影后，我们得到：

$$T^{3D} = \text{MLP}_{\text{proj}}(\gamma(X)) \in R^{N_{pts} \times C}. \quad (4)$$

从输入图像 $I \in RH \times W \times 3$ 出发，冻结体编码器 E_{body} 生成特征矩阵 $RN_{\text{body}} \times C_b$ ，该矩阵随后被投影至

$$T^{\text{body}} = \text{MLP}_{\text{proj}}(E_{\text{body}}(I)) \in RN_{\text{body}} \times C. \quad (5)$$

为保留精细的面部细节，冻结的多尺度头部编码器可生成 $\{E_{\text{head}}^{(d)}(I) \in R^{N_{\text{head}}^{(d)} \times C_h \times d} | d=1\}$ ；融合模块将它们聚合为

$$T^{\text{head}} = F_{\text{fusion}}(\{E_{\text{head}}^{(d)}(I) | d=1\}) \in RN_{\text{head}} \times C, \quad (6)$$

其中 N_{head} 表示融合后的令牌数量。

预测：多模态身体-头部变换器 (MBHT) 融合三个标记集后，解码器可预测标准高斯分布及骨骼参数：

$$(\chi, \{wik\}, \{Tk(\cdot)\}) = \text{解码器}(\text{MBHT}(T^{3D}, T^{\text{body}}, T^{\text{head}})). \quad (7)$$

此处， χ 包含 N 个高斯分布， $wik \geq 0$ 且 $\sum wik = 1$ ；而 $Tk(\theta) \in \text{SE}(3)$ 是在规范空间中定义的刚性骨骼变换。

具体而言，标准高斯集是

$$\chi = \left\{ (p_i, \sigma_i, r_i, \rho_i, f_i) \right\}_{i=1}^N, \quad (8)$$

其中 $\pi \in R^3$ (质心)、 $\sigma_i \in R^3$ (各向异性尺度)、 $r_i \in S^3$ (单位四元数)、 $\rho_i \in [0,1]$ (不透明度)，以及 $f_i \in R_k$ (SH 外观)。三维协方差矩阵为

$$\Sigma_i = R(r_i) \text{diag}(\sigma_i^2) R(r_i), \quad R(r_i) \in \text{SO}(3), \quad (9)$$

其中 $R(\cdot)$ 表示四元数到旋转的映射关系。为实现动画效果，该模块提供了一个兼容 SMPL-X 的骨骼模型，其混合权重 wik 和骨骼变换 $Tk(\theta) \in \text{SE}(3)$ 用于定义姿态 θ 。

姿态条件化 (运行时)：给定来自 VR 前端的每帧姿态参数 θ ，我们使用具有高斯权重 wik 的线性混合蒙皮 (LBS) 技术，将标准高斯集 χ 变形为姿态集 $\chi' = \text{LBS}(\chi, \theta)$ 。我们将高斯分量 i 的 LBS 输出定义为 $(p'_i, \Sigma'_i) = \text{LBS}(p_i, \Sigma_i; \theta)$ 。具体而言，令 $Tk(\theta) = (R_k(\theta), tk(\theta))$ 表示骨骼 k 的姿态依赖性变换，而 $R_i(\theta)$ 表示诱导的混合旋转，我们有

$$p'_i = \sum_k w_{ik} (R_k(\theta) p_i + tk(\theta)), \quad (10)$$

$$\Sigma'_i = R_i(\theta) \Sigma_i R_i(\theta). \quad (11)$$

分布与合成：我们采用标准的三维高斯分布光栅化技术 [21] 来渲染姿态化的高斯分布。

$$\text{即} = \text{渲染}(\text{LBS}(\chi, \theta), \mathbf{K}, \mathbf{TCW}, e), \quad e \in \{L, R\}, \quad (12)$$

其中 $\text{LBS}(\chi, \theta)$ 表示上述 LBS 步骤生成的高斯集合，而 $(\mathbf{K}, \mathbf{TCW}, e)$ 是每只眼睛的相机参数。在双目批量处理方法中，每个高斯属性会被一次性生成，并在同一批次内同时用于左和右眼视图；而每只眼睛的投影计算和视图依赖性着色则在同一处理流程中分别完成。

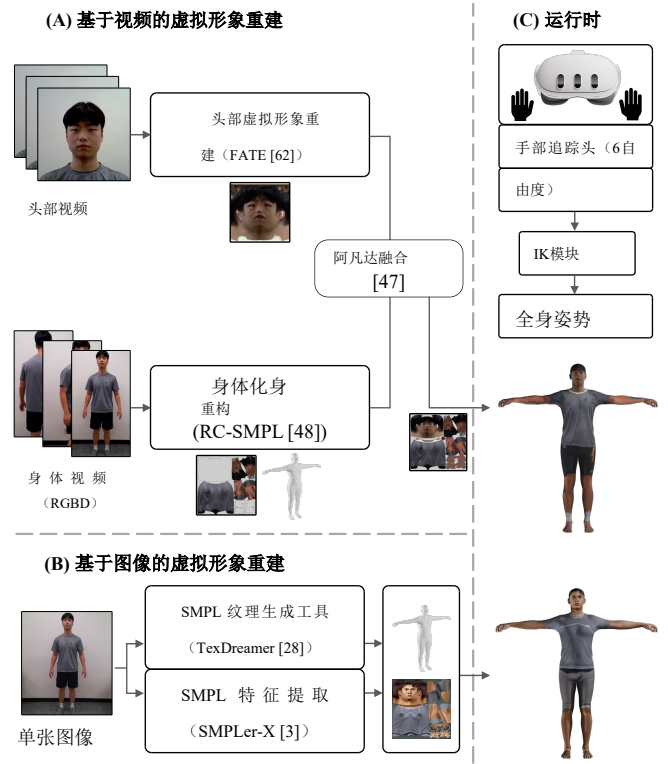


图 3：基线条件。(A) 基于视频的重建：根据先前研究 [47、48、62] 实现。(B) 基于图像的重建：根据先前研究 [3、28] 实现。(C) 运行流程：IK 模块通过头部 6DoF 位姿和手部追踪信号估算全身姿态，以实时动画化虚拟角色。

编码及传输至头戴式显示器 (HMD)。渲染后的帧被转换为 8 位 sRGB 格式，并以固定质量进行 JPEG 压缩；左侧/右侧的有效载荷连同 frame_id 通过单一响应包传输，并直接送达 HMD。这种无状态的逐帧设计允许接收端确定性丢弃延迟帧。

GA 后端会为 3DGS 虚拟形象添加一个 alpha 通道，以便在包含 3DGS 场景/背景时支持可选的虚拟形象合成功能。在本研究中，我们禁用了背景合成功能，并采用黑色背景以最大限度降低背景干扰，从而将评估重点集中于虚拟形象本身；为此，我们仅传输 RGB 格式的 JPEG 图像帧以降低带宽消耗。

算法概述。

1. 从 VR 前端接收 $\{\text{frame_id}, (\mathbf{TCW}, e, \mathbf{K}), \theta\}$ 。
2. 构建单批次立体渲染环境，并对标准 3DGS 虚拟形象应用姿态条件化处理，根据 θ 对高斯基元进行变形。
3. 使用 *双眼批量处理* 执行一次批量分割操作，同时渲染左眼和右眼图像 $[L, R]$ 。
4. 对两个渲染完成的眼部缓冲区进行 JPEG 压缩，并将其与 frame_id 一同打包。
5. 将压缩后的响应直接流式传输回头戴式显示器 (HMD) 进行显示。

实施细节。GA Backend 已实现于 Python 3.9⁴、PyTorch 2.3.0⁵ 及 PyTorch3D⁶。渲染过程实时运行于单块 NVIDIA RTX 4090 GPU 上。

⁴<https://www.python.org/>

⁵<https://pytorch.org/>

⁶<https://pytorch3d.org/>



Fig. 4: (A) User study setup with participants performing poses from pre-recorded voice instructions. (B) Avatar type factor with two levels: *self* (left, resembling the participant) and *other* (right, resembling another person).

4 EXPERIMENTS

4.1 Baselines

To evaluate our system, we compare against mesh-based full-body avatar reconstruction methods using the SMPL-X [38] model. Figure 3 illustrates the baseline systems constructed for comparison: (A) video-based mesh avatar reconstruction, and (B) image-based mesh avatar reconstruction. For 3DGS avatar generation in our system, VRGaussianAvatar (*Ours*), a frontal image of each subject in an A-pose with arms slightly extended was captured. This single image served as the input for generating a full-body 3DGS avatar.

In the case of video-based mesh avatar reconstruction (*video*), we separately utilized head and body videos. For the head, we recorded a one-minute video of the Facial Action Coding System (FACS) [8]-based facial expressions and reconstructed it with FATE [62] to obtain a textured FLAME [26] model. For the body, we captured RGBD videos using the Azure Kinect DK⁷ and reconstructed the textured SMPL-X model with RC-SMPL [48]. The head and body textures were subsequently merged using the method of [47].

For image-based mesh avatar reconstruction (*image*), we used the same single input image employed in VRGaussianAvatar. We adopted the off-the-shelf SMPL-X inference model SMPLer-X [3] for mesh recovery, and generated textures with TexDremer [28], an image-based SMPL-X texture generator.

Experimental Details. All baselines and the VRGaussianAvatar runtime were executed on a desktop PC equipped with an Intel Core i9-14900K CPU, an NVIDIA RTX 4090 GPU, and 64 GB of RAM. For avatar reconstruction, both the image-based mesh baseline and our system’s Gaussian Avatar Module were run on the same desktop PC as the runtime environment. The image-based baseline required approximately 30 seconds for SMPL-X parameter estimation and 19 seconds for texture generation. Our Gaussian Avatar Module reconstructed a full-body 3D Gaussian Avatar within 10 seconds. For the video-based mesh baseline, body reconstruction with RC-SMPL took about 30 seconds, and avatar integration required an additional 2 seconds. Facial reconstruction with FATE was performed on a separate workstation equipped with an NVIDIA A40 GPU and required about 12 hours.

4.2 User Study

We conducted a human-subject study to evaluate whether our system can effectively generate realistic user avatars for VR. The evaluation considered two perspectives: (1) whether a self-resembling avatar is perceived as one’s own, and (2) whether an avatar resembling another person can be accepted as one’s own. Based on these perspectives, we compared our system with two baseline methods.

Study Design and Task.

The experimental factors and conditions were organized as follows. The first factor was avatar type with two levels: (1) self-representative avatar (*self*) and (2) other-representative avatar (*other*). For the *other* avatar type, we used avatars of non-participant models matched in gender, ethnicity, and similar age to the participants (Figure 4 (B)). The second factor was avatar reconstruction method with three conditions related to the generation process: (1) *Ours* (VRGaussianAvatar), (2) *video* (video-based avatar reconstruction) [47, 48, 62], and (3) *image* (image-based avatar reconstruction) [3, 28] methods. Both of the avatar type and method factors were within-subject factors, our experiment employed a 2×3 within-subject factorial design. To compare method conditions under the self and other perspectives, avatar type was counterbalanced across participants, and the three method conditions within each type were also counterbalanced using a Latin Square design.

To evaluate the appearance and movements of avatars in VR, we adapted a commonly used avatar control task from prior studies assessing virtual embodiment [7, 44, 54]. In this task, participants wearing an HMD experienced their avatars in mirror mode and were instructed through audio cues to make specific body poses, while observing how these poses were reflected in their avatar [44]. We chose the mirror-based observation (rather than a purely first-person view) because our primary goal was to let participants directly inspect the reconstructed avatar’s overall appearance and motion quality as a self-representation. Building on the previous work [7], which extended the original task [44] by adding side-view observation instructions, we designed the task to better fit our study context and to ensure that participants could sufficiently observe the avatars. For instance, participants were encouraged to freely explore their avatars (e.g., “Move your arms and take any posture you want.”) and were also given additional instructions to observe their avatars not only from the front, but also from the left, right, and rear views.

Dependent Variables. In this study, the Sense of Embodiment (SoE) was measured as the primary dependent variable, as the study aimed to evaluate whether our realistic avatars’ appearance and movements conveyed appropriate body representation and perception [2, 44]. Based on the previous studies that explored overall perception and embodiment of avatars in VR [30, 31, 33, 54, 58], we adopted two representative measurements: (1) the Virtual Embodiment Questionnaire (VEQ) by Roth and Latoschik [44] and (2) VEQ+ by Fiedler et al. [11]. The VEQ, which is specifically designed to assess SoE in VR environments, consists of three latent factors. First, Virtual Body Ownership (VBO) refers to the perception of a virtual body as one’s own. Second, Agency (AG) denotes the perception of control over the virtual body, and the last Change (CH) captures alterations in the perceived body schema. The VEQ+ extends this framework by incorporating two additional dimensions: Self-location (the feeling of being located at the position of one’s body) and Self-identification (related to personalization and resemblance). Under these dimensions, VEQ+ has three subscales—Self-Location (SL), Self-Attribution (SA), and Self-Similarity (SS). Both the VEQ and VEQ+ comprise 12 items, for a total of 24 items.

The Virtual Human Plausibility Questionnaire (VHPQ) measures the extent to which a virtual avatar’s appearance and behavior are perceived as plausible, coherent, and consistent with the VR environment [29]. Considering the objective and setup of our system evaluation, we only used the Appearance and Behavior Plausibility (ABP) subscale of the VHPQ, which consisted of six items. In addition, following earlier research [44], we also included three self-reported items to more di-

⁷<https://learn.microsoft.com/en-us/azure/kinect-dk/>



图4: (A) 用户研究实验设置, 参与者根据预先录制的语音指令完成动作。(B) 虚拟形象类型因素包含两个水平: 自我(左图, 与参与者相似)和其他(右图, 与另一个人相似)。

4 实验

4.1 基线

为评估我们的系统, 我们将其与基于网格的全身虚拟形象重建方法(采用 SMPL-X [38]模型)进行对比。图3展示了用于比较的基准系统: (A) 基于视频的网格虚拟形象重建方法; (B) 基于图像的网格虚拟形象重建方法。在本系统的3DGS虚拟形象生成过程中, 我们采集了每位受试者呈A字姿势、双臂略微伸展的正面图像——即VR Gaussian Avatar(本系统)。该单张图像被用作生成全身3DGS虚拟形象的输入数据。

在基于视频的网格化身重建(视频)过程中, 我们分别使用了头部和身体的视频。对于头部, 我们录制了一分钟基于面部动作编码系统(FACS) [8]的面部表情视频, 并利用FATE[62]对其进行重建, 从而获得纹理化的火焰模型[26]; 对于身体, 则使用Azure Kinect DK⁷采集 RGBD 视频, 并通过RC-SMPL [48]重建 SMPL-X 纹理化模型。随后采用[47]所述方法将头部与身体的纹理进行融合。

对于基于图像的网格化身重建(图像), 我们采用了与VR-Gaussian Avatar相同的单一输入图像。我们使用现成的 SMPL-X 推理模型 SMPLer-X[3]进行网格重建, 并通过基于图像的 SMPL-X 纹理生成器 TexDreamer[28]生成纹理。

实验细节。所有基线方法及VR Gaussian Avatar运行环境均在一台配备Intel Core i9-14900K处理器、NVIDIA RTX 4090显卡和64 GB内存的台式电脑上运行。在虚拟化身重建方面, 基于图像的网格基线方法与本系统的Gaussian Avatar模块均在同一台台式电脑上运行。基于图像的基线方法完成 SMPL-X 参数估计需约30秒, 纹理生成耗时19秒; 而我们的Gaussian Avatar模块在10秒内即可完成全身3D高斯虚拟化身的重建。对于基于视频的网格基线方法, 使用RC-SMPL 进行身体重建约需30秒, 虚拟化身整合还需额外2秒; 而采用FATE进行面部重建则在配备 NVIDIA A40 显卡的独立工作站上完成, 耗时约12小时。

4.2 用户研究

我们开展了一项人体受试者研究, 旨在评估我们的系统能否有效生成适用于虚拟现实(VR)的逼真用户虚拟形象。评估从两个角度进行: (1)具有自身特征的虚拟形象是否会被感知为本人; (2)与他人相似的虚拟形象是否能被接受为本人。基于这些评估标准, 我们将本系统与两种基线方法进行了比较。

研究设计与任务。

实验因素及.....
实验条件设置如下。第一个因素是虚拟形象类型, 包含两个层级: (1)自我代表型虚拟形象(self); (2)他人代表型虚拟形象(other)。对于other类型虚拟形象, 我们采用了与参与者性别、种族及年龄相匹配的非参与者模型虚拟形象(图4(B))。第二个因素是虚拟形象重建方法, 包含与生成过程相关的三种条件: (1)Ours (VR高斯虚拟形象); (2)video (基于视频的虚拟形象重建) [47, 48, 62]; (3)image。

(基于图像的虚拟形象重建) [3, 28] 方法。虚拟形象类型与方法因素均为被试内因素; 本实验采用 2×3 被试内析因设计。为从自我与他人视角比较不同方法条件, 虚拟形象类型在参与者间进行了平衡设计, 且每种类型下的三种方法条件也采用拉丁方设计进行了平衡。

为评估虚拟现实中的虚拟形象的外观与动作, 我们借鉴了先前评估虚拟具身性的研究中常用的虚拟形象控制任务[7, 44, 54]。在该任务中, 佩戴头戴式显示器(HMD)的参与者以镜像模式观察自己的虚拟形象, 并通过音频提示完成特定的身体姿势动作, 同时观察这些姿势如何在虚拟形象上呈现[44]。我们选择基于镜像的观察方式(而非纯粹的第一人称视角), 是因为我们的主要目标是让参与者能够直接审视重构后的虚拟形象的整体外观与动作质量——这被视为一种自我表征。基于先前研究[7] (该研究通过增加侧视观察指令扩展了原始任务[44]), 我们设计了更符合本研究背景的任务方案, 以确保参与者能充分观察虚拟形象。例如, 我们鼓励参与者自由探索自己的虚拟形象(例如: “移动手臂, 摆出任意姿势。”)并额外要求参与者不仅从前视角观察自己的虚拟形象, 还需从左、右及后方视角进行观察。

因变量。本研究将具身感(SoE)作为主要因变量进行测量, 因为研究旨在评估我们设计的逼真虚拟形象的外观与动作是否能传递恰当的身体表征与感知[2, 44]。基于先前关于虚拟现实环境中虚拟形象整体感知与具身性的研究[30, 31, 33, 54, 58], 我们采用了两项代表性测量工具: (1)Roth与Latoschik开发的虚拟具身问卷(VEQ) [44]; (2)Fiedler等人设计的 VEQ+版本。[11]该VEQ量表专为评估虚拟现实环境中的自我体验(SoE)而设计, 包含三个潜在因子: 首先, 虚拟身体所有权(VBO)指个体将虚拟身体视为自身的一部分; 其次, 能动性(AG)体现对虚拟身体的控制感知; 最后, 变化(CH)反映感知到的身体图式发生的变化。VEQ+量表在此框架基础上增加了两个维度: 自我定位(即身处自身身体位置的感觉)和自我认同(涉及个性化与相似性)。基于这两个维度, VEQ+包含三个子量表——自我定位(SL)、自我归因(SA)和自我相似性(SS)。VEQ与VEQ+各包含12个项目, 共计24个项目。

虚拟人类可信度问卷(VHPQ)用于评估虚拟化身的外貌与行为在多大程度上被认为具有可信度、逻辑连贯性, 并与虚拟现实环境相契合[29]。鉴于本系统评估的目标与设计的要求, 我们仅采用了VHPQ中的“外貌与行为可信度”(ABP)子量表, 该子量表包含六个条目。此外, 参照先前研究[44], 我们还加入了三项自我报告条目以增强评估的全面性。

⁷<https://learn.microsoft.com/en-us/azure/kinect-dk/>

rectly compare the realism, naturalness, and visuomotor synchrony of avatar motion behavior: “*The virtual avatar’s movements were realistic*,” “*The virtual avatar’s movements were naturalistic*,” and “*The virtual avatar’s movements were in synchrony with my own movements*.” All questionnaire measures were assessed on a 7-point Likert scale (1: Strongly disagree – 7: Strongly agree). For the post-experiment interview, we designed open-ended questions with reference to Menzel et al. [33] to collect qualitative feedback on avatar fidelity, suitability, and realism when observing both *self* and *other* avatar types. Participants were also asked about their subjective opinions on differences between avatars and their preferences.

Procedure. This study was approved by the Institutional Review Board (IRB). Each participant attended two sessions. The first session was for data acquisition for the *video* method, where participants’ facial data were scanned (pre-processing required over 12 hours); They also received an explanation of the study, provided informed consent, and completed a demographic questionnaire. The second session, conducted at least one day later, was the main experiment. At the beginning, participants’ bodies were scanned to generate avatars for the *Ours* and *image* conditions, while the *video* avatars were generated by combining the pre-processed facial model with the newly scanned body.

After avatar generation, participants wore an HMD and performed the avatar control task while standing in an empty experimental space (Figure 4 (A)). Following pre-recorded voice instructions, they assumed various body poses and observed their virtual avatars mirroring their poses in real-time. After each method condition, they removed the HMD and completed the post-task questionnaire. Participants first completed the three method conditions (*Ours*, *video*, *image*) arranged in a balanced order under one avatar type, and then repeated the sequence under the other type, resulting in six evaluations in total. Finally, a post-experiment interview was conducted to collect qualitative feedback on their experiences with each method under both *self* and *other* conditions.

Participants. We recruited 24 participants (12 male, 12 female) through the campus website. Their ages ranged from 22 to 34 years ($M = 27.67$, $SD = 3.17$). Regarding prior experience with avatar-mediated applications (e.g., social networking services, games), 17 participants reported using such applications fewer than five times (70.83%), whereas seven participants reported using them five times or more (29.17%). In contrast, most participants (20, 83.3%) indicated that they had used AR/VR technologies and wearable devices more than five times, while only four participants reported fewer than five times of AR/VR experience (16.67%).

4.3 Quantitative Experiments

We conducted a quantitative evaluation of the Gaussian Avatar Module, a core component of VRGaussianAvatar responsible for avatar representation. For comparison, we employed video-based [47, 48, 62] and image-based [3, 28] avatar reconstruction methods that are also applicable in VR, consistent with our user study setup. We also measured the rendering speed at different resolutions to assess the contribution of Binocular Batching for real-time rendering.

Datasets. We recruited 24 participants (12 male and 12 female) for evaluation. For both VRGaussianAvatar and the image-based baseline, a single source image was captured with the subject standing in an A-pose (arms lowered and slightly apart from the body). For the video-based baseline, we recorded videos for facial and body reconstruction. Facial reconstruction used a one-minute RGB video containing diverse expressions based on the FACS [8], while body reconstruction relied on a 30-second RGBD video capturing both the frontal and rear views of the subject. For fair evaluation, rendered results from each method were cropped to the full-body and compared against the corresponding ground-truth images.

Metrics. We employed four widely used metrics for evaluating avatar reconstruction quality: L1 distance, LPIPS [63], PSNR [15], and

SSIM [55]. L1 distance measures pixel-wise absolute error between the rendered image and ground truth. LPIPS (Learned Perceptual Image Patch Similarity) assesses perceptual similarity based on deep neural network features, better reflecting human visual perception. PSNR (Peak Signal-to-Noise Ratio) quantifies reconstruction quality in terms of signal fidelity, with higher values indicating less distortion. SSIM (Structural Similarity Index) evaluates structural consistency and perceptual quality, particularly in terms of luminance, contrast and structure.

5 RESULTS

5.1 User Study Analysis

For subjective measures, a two-way repeated measures ANOVA with the Aligned Rank Transform (ART) was used for non-parametric factorial analysis ($\alpha = .05$) [57]. The ART method has been widely utilized in human-subject research analysis because it is specialized in multi-factorial non-parametric analysis and accommodates ordinal and continuous variables [57]. All post-hoc comparisons were conducted using the ART-C procedure [9] with Bonferroni corrected p-values. The internal consistency among Likert items was examined by the Cronbach’s alpha reliability coefficient.

5.1.1 Virtual Embodiment

VEQ. The results for the VEQ and its subscales are shown in Figure 6. The VEQ score, which aggregates all three subscales, showed a reliable consistency ($\alpha_{VEQ} = .825$), and the three subscales also showed an acceptable value ($\alpha_{VBO} = .861$; $\alpha_{AG} = .837$; $\alpha_{CH} = .890$). We found a main effect of avatar type ($F_{1,23} = 8.464$, $p = .008$, $\eta_p^2 = .269$), with *self* rated higher than *other* avatar types. There were no significant main effect of method ($F_{2,46} = 1.082$, $p = .348$, $\eta_p^2 = .045$), nor interaction effect between the two factors ($F_{2,46} = 2.493$, $p = .094$, $\eta_p^2 = .098$).

For VBO, significant main effects were found for avatar type ($F_{1,23} = 15.064$, $p < .001$, $\eta_p^2 = .396$) and method ($F_{2,46} = 21.903$, $p < .001$, $\eta_p^2 = .488$). Post-hoc tests indicated that *self* avatars were rated higher than *other*, and *Ours* method induced higher ownership than both *video* and *image* (all $p < .001$), while *video* and *image* did not differ. No interaction effect was observed ($F_{2,46} = 1.419$, $p = .252$). AG showed significant main effects of avatar type ($F_{1,23} = 4.323$, $p = .049$, $\eta_p^2 = .158$) and method ($F_{2,46} = 6.041$, $p = .005$, $\eta_p^2 = .208$). Post-hoc comparisons revealed that *self* avatars scored higher than *other* avatars, and that *Ours* conveyed higher agency than both *video* ($p = .031$) and *image* ($p = .006$), while *video* and *image* were not different. The interaction was not significant ($F_{2,46} = 0.841$, $p = .438$). The CH score, with lower values reflecting less perceived change from one’s own body and thus a better representation of the actual user, showed a significant main effect of method ($F_{2,46} = 13.728$, $p < .001$, $\eta_p^2 = .374$), but no main effect of avatar type. Pairwise comparisons showed that *Ours* yielded lower CH scores than both *video* ($p = .001$) and *image* ($p < .001$), whereas *video* and *image* did not differ. In addition, a significant interaction effect was observed between two factors ($F_{2,46} = 7.932$, $p = .001$, $\eta_p^2 = .256$). Follow-up analyses showed that in the *self* condition, *Ours* produced lower CH scores than both *video* and *image* (all $p < .001$), while in the *other* condition, *Ours* was only lower than *image* ($p = .024$).

VEQ+. VEQ+ ($\alpha_{VEQ+} = .937$) and its subscales all showed an acceptable Cronbach’s alpha ($\alpha_{SL} = .765$; $\alpha_{SA} = .889$; $\alpha_{SS} = .965$). The results of VEQ+ are also in Figure 6. The overall VEQ+ score revealed significant main effects of avatar type ($F_{1,23} = 57.715$, $p < .001$, $\eta_p^2 = .715$) and method ($F_{2,46} = 49.487$, $p < .001$, $\eta_p^2 = .683$), and a significant interaction effect ($F_{2,46} = 16.155$, $p < .001$, $\eta_p^2 = .413$). Post-hoc tests showed that *self* avatars scored higher than *other*, and *Ours* was significantly higher than both *video* and *image* (all $p < .001$). Also, *video* was rated higher than *image* ($p < .001$). Within the *self* condition, *Ours* was significantly higher than both *video* and *image* (all $p < .001$), and *video* was rated higher than *image* ($p = .005$). Within the *other* condition, *Ours* again scored significantly higher than

直接比较虚拟化身运动行为的真实感、自然度及视觉运动同步性：“虚拟化身的动作非常逼真”，“虚拟化身的动作十分自然”，以及“虚拟化身的动作与我的实际动作完全同步”。所有问卷测量指标均采用7点李克特量表（1：强烈不同意 - 7：强烈同意）进行评估。在实验后的访谈中，我们参照Menzel等人的研究设计了开放式问题。[33]旨在收集参与者在观察自身及其他类型虚拟形象时，对虚拟形象逼真度、适用性及真实感的定性反馈。同时，研究人员还询问了参与者对虚拟形象间差异的主观看法及其偏好。

研究流程。本研究已获得机构审查委员会（IRB）的批准。每位参与者均参加了两次实验环节。首次实验环节采用视频方法进行数据采集：参与者面部数据被扫描（预处理耗时超过12小时）；同时他们接受了研究说明、签署了知情同意书并完成了人口统计问卷。第二次实验环节至少在一天后进行，为正式实验部分。实验开始时，首先对参与者的身体进行扫描以生成我们与图像条件下的虚拟形象；而视频虚拟形象则是通过将预处理后的面部模型与新扫描的身体数据相结合生成的。

在生成虚拟形象后，参与者佩戴头戴式显示器（HMD），站在空旷的实验空间中执行虚拟形象控制任务（图4(A)）。根据预先录制的语音指令，他们采取多种身体姿势，并实时观察自己的虚拟形象同步呈现相应姿态。每种方法条件结束后，参与者摘下HMD并完成任务后问卷调查。首先，参与者在虚拟形象类型下按平衡顺序完成三种方法条件（本研究方法、视频、图像），随后在另一种类型下重复相同流程，共计完成六次评估。最后通过实验后访谈收集参与者在自我与他人两种情境下对各方法体验的定性反馈。

参与者。我们通过校园网站招募了24名参与者（12名男性，12名女性），年龄范围为22至34岁（平均值 = 27.67，标准差 = 3.17）。关于先前使用虚拟化身中介应用（如社交网络服务、游戏）的经验，17名参与者表示使用次数少于五次（70.83%），而7名参与者表示使用次数达到五次或以上（29.17%）。相比之下，大多数参与者（20人，占83.3%）表示曾使用AR/VR技术及可穿戴设备超过五次，仅有四名参与者报告的AR/VR使用次数少于五次（16.67%）。

4.3 定量实验

我们对Gaussian Avatar Module进行了定量评估，该模块是VR Gaussian Avatar的核心组件，负责虚拟形象的呈现。为便于比较，我们采用了基于视频的[47, 48, 62]和基于图像的[3, 28]虚拟形象重建方法，这些方法同样适用于虚拟现实场景，且与我们的用户研究设置一致。同时，我们还测量了不同分辨率下的渲染速度，以评估双目分批处理技术对实时渲染的贡献。

数据集。我们招募了24名参与者（12名男性和12名女性）进行评估。对于VR Gaussian Avatar方法和基于图像的基线方法，均拍摄了受试者以A位姿势（双臂下垂并与身体略微分开）站立的单张源图像；而对于基于视频的基线方法，则录制了用于面部和身体重建的视频：面部重建使用了一分钟的RGB视频（视频包含基于FACS[8]的多种表情），而身体重建则采用一段30秒的RGBD视频，该视频同时捕捉了受试者的正面和背面视角。为确保评估公平性，每种方法生成的渲染结果均被裁剪为全身尺寸，并与相应的真实图像进行对比。

评估指标。我们采用了四种广泛使用的指标来评估虚拟形象重建质量：L1距离、LPIPS [63]、PSNR [15]。

SSIM [55]。L1距离用于衡量渲染图像与真实图像在像素层面的绝对误差；LPIPS（学习感知图像块相似性）基于深度神经网络特征评估感知相似性，能更准确反映人类视觉感知；PSNR（峰值信噪比）通过信号保真度量化重建质量，数值越高表明失真越小；SSIM（结构相似性指数）则评估结构一致性与感知质量，尤其关注亮度、对比度及结构特征。

5 结果

5.1 用户研究分析

对于主观测量指标，采用经对齐秩变换（ART）处理的双因素重复测量方差分析进行非参数析因分析（ $\alpha = 0.05$ ）[57]。ART方法因其专精于多因素非参数分析且能处理有序变量和连续变量，已在人类受试者研究分析中得到广泛应用[57]。所有事后比较均采用ART-C程序[9]进行，并采用Bonferroni校正后的p值。Likert量表条目的内部一致性通过Cronbach's α 信度系数进行评估。

5.1.1 虚拟化身

VEQ。VEQ及其各子量表的结果如图6所示。综合三个子量表的VEQ得分显示出良好的一致性（ $\alpha_{VEQ} = .825$ ），三个子量表的信度值也均处于可接受范围（ $\alpha_{VBO} = .861$ ； $\alpha_{AG} = .837$ ； $\alpha_{CH} = .890$ ）。我们发现虚拟形象类型存在主效应（ $F_{1, 23} = 8.464$ ， $p = .008$ ， $\eta^2_p = .269$ ），其中自我类型的评分高于其他虚拟形象类型。方法因素未呈现显著主效应（ $F_{2, 46} = 1.082$ ， $p = .348$ ， $\eta^2_p = .045$ ），两个因素之间亦无交互效应（ $F_{2, 46} = 2.493$ ， $p = .094$ ， $\eta^2_p = .098$ ）。

例如，发现虚拟形象类型（ $F_{1, 23} = 15.064$ ， $p < .001$ ， $\eta^2_p = .396$ ）和方法（ $F_{2, 46} = 21.903$ ， $p < .001$ ， $\eta^2_p = .488$ ）存在显著的主效应。事后检验表明：自我虚拟形象的评分高于其他虚拟形象；我们的方法产生的归属感高于视频和图像方法（所有 $p < .001$ ），而视频与图像之间无显著差异。未观察到交互效应（ $F_{2, 46} = 1.419$ ， $p = .252$ ）。AG分析显示虚拟形象类型（ $F_{1, 23} = 4.323$ ， $p = .049$ ， $\eta^2_p = .158$ ）与方法（ $F_{2, 46} = 6.041$ ， $p = .005$ ， $\eta^2_p = .208$ ）均存在显著主效应。事后比较显示：自我化身的得分高于其他化身；我们的化身所传递的自主性高于视频（ $p = .031$ ）和图像（ $p = .006$ ），而视频与图像之间无显著差异。交互效应不显著（ $F_{2, 46} = 0.841$ ， $p = .438$ ）。CH得分（数值越低表示对自身身体变化的感知越小，从而更能反映真实用户）显示出方法的显著主效应（ $F_{2, 46} = 13.728$ ， $p < .001$ ， $\eta^2_p = .374$ ），但化身类型无主效应。成对比较表明：我们的化身的CH得分低于视频（ $p = .001$ ）和图像（ $p < .001$ ），而视频与图像之间无差异。此外，两个因素间存在显著的交互效应（ $F_{2, 46} = 7.932$ ， $p = .001$ ， $\eta^2_p = .256$ ）。后续分析表明，在自我条件下，我们的表现得出的CH分数低于视频和图像（所有 $p < .001$ ）；而在其他条件下，我们的分数仅低于图像（ $p = .024$ ）。

VEQ+。VEQ+（ $\alpha_{VEQ+} = 0.937$ ）及其各子量表均显示出可接受的克隆巴赫 α 系数（ $\alpha_{SL} = 0.765$ ； $\alpha_{SA} = 0.889$ ； $\alpha_{SS} = 0.965$ ）。VEQ+的结果亦展示于图6。总体VEQ+得分显示出虚拟形象类型（ $F_{1, 23} = 57.715$ ， $p < .001$ ， $\eta^2_p = 0.715$ ）与方法（ $F_{2, 46} = 49.487$ ， $p < .001$ ， $\eta^2_p = 0.683$ ）的显著主效应，以及显著的交互效应（ $F_{2, 46} = 16.155$ ， $p < .001$ ， $\eta^2_p = 0.413$ ）。事后检验表明：自我虚拟形象得分高于其他虚拟形象；我们的虚拟形象得分显著高于视频与图像虚拟形象（所有 $p < .001$ ）；此外，视频虚拟形象的评分也高于图像虚拟形象（ $p < .001$ ）。在自我条件下，我们的评分显著高于视频和图像（所有 $p < .001$ ）；而视频的评分也高于图像（ $p = .005$ ）。在其他条件下，我们的评分再次显著高于

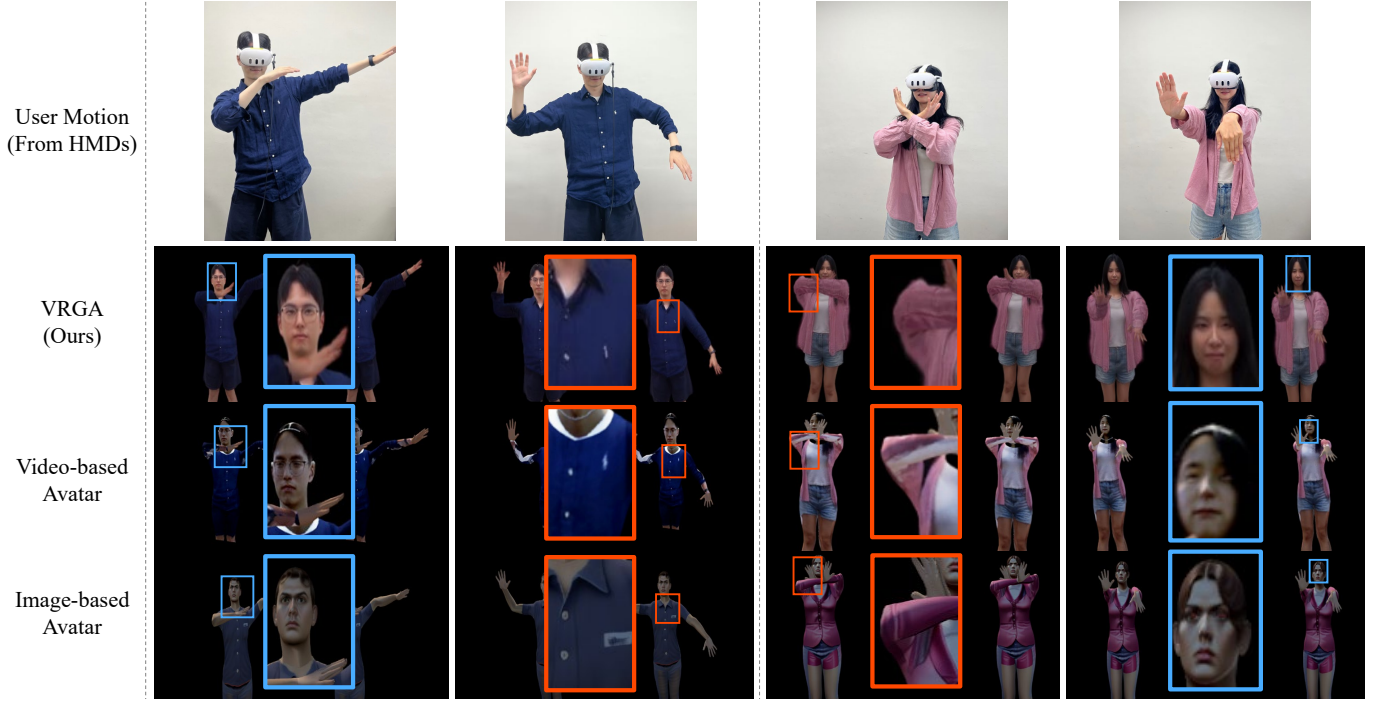


Fig. 5: Qualitative comparison of VRGaussianAvatar with baseline avatar representations under identical HMD-only control. Compared to baselines, VRGaussianAvatar typically preserves sharper appearance details in close-up regions, including head/face (sky-blue boxes) and clothing (orange boxes). All methods are rendered stereoscopically in real time, enabling interactive avatar control in VR.

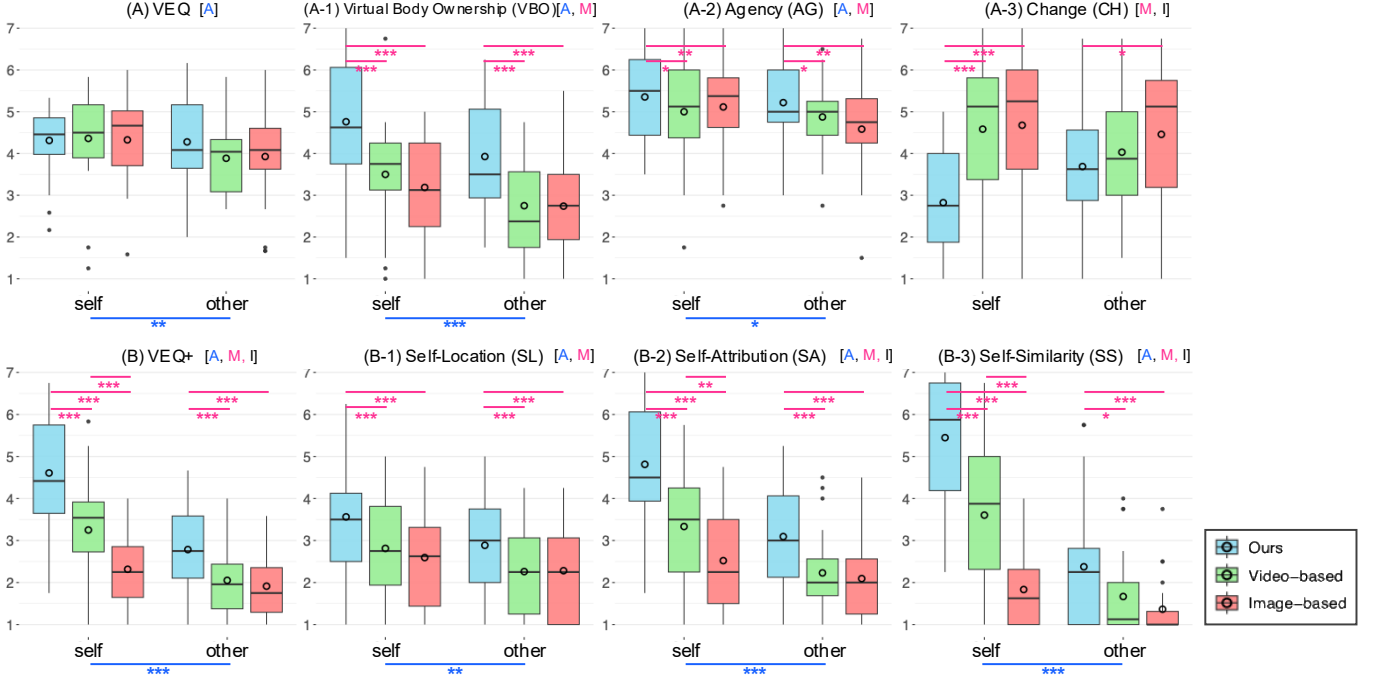


Fig. 6: Results for (A)–(A-3) Virtual Embodiment Questionnaire and its subscales; (B)–(B-3) VEQ+ and its subscales. (A and M: significant main effect of avatar type and method, respectively; I: significant interaction effect between avatar type and method)

video and image ($p < .001$), while the difference between video and image was not significant.

SL showed significant main effects of avatar type ($F_{1,23} = 9.389$, $p = .005$, $\eta_p^2 = .290$) and method ($F_{2,46} = 12.948$, $p < .001$, $\eta_p^2 = .360$). Post-hoc comparisons indicated that *self* avatars were rated higher than *other* avatars, and *Ours* showed significantly higher SL

than *video* ($p = .001$) and *image* ($p < .001$), whereas *video* and *image* did not differ. No interaction effect was found. SA revealed significant main effects of avatar type ($F_{1,23} = 45.904$, $p < .001$, $\eta_p^2 = .666$) and method ($F_{2,46} = 44.525$, $p < .001$, $\eta_p^2 = .659$), as well as a significant interaction effect ($F_{2,46} = 10.653$, $p < .001$, $\eta_p^2 = .317$). Post-hoc analysis showed that *self* type was higher than *other* type; *Ours* method

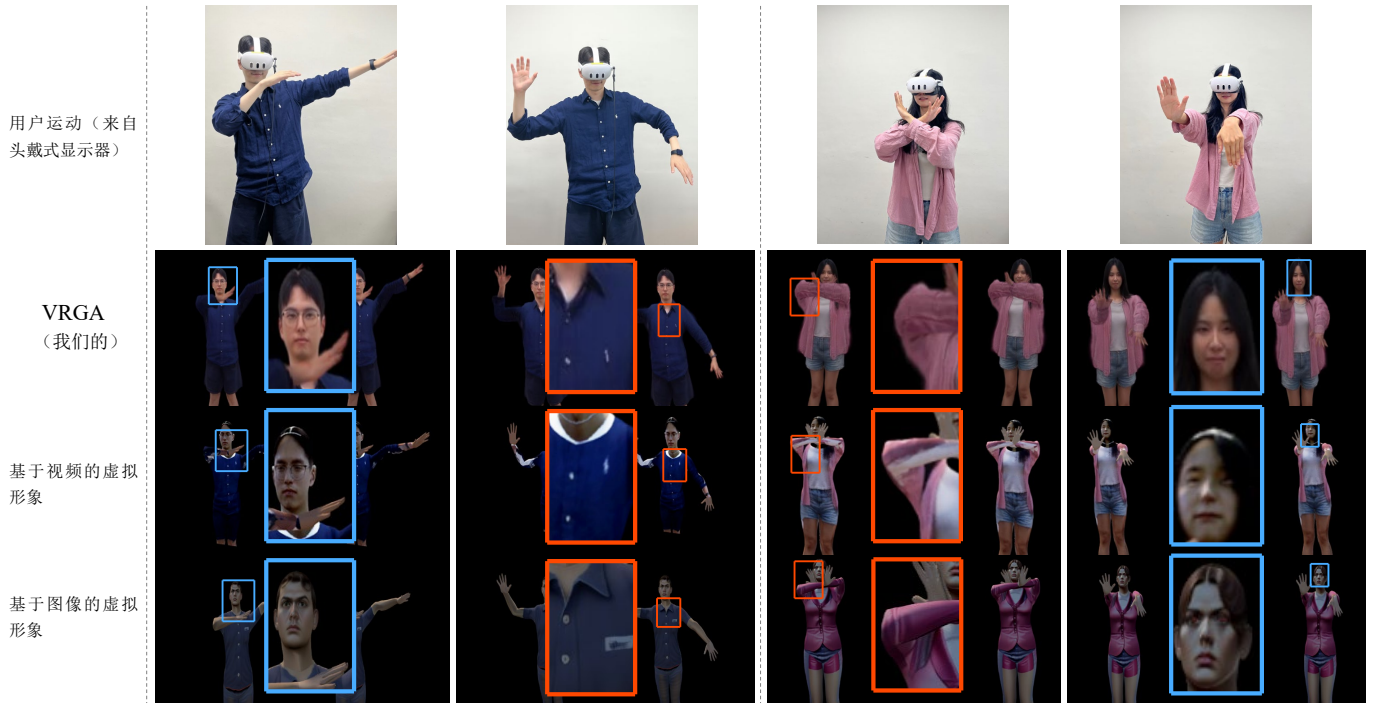


图5：在仅使用头戴式显示器（HMD）的相同控制条件下，VR Gaussian Avatar与基线虚拟形象表示的定性比较。与基线方法相比，VR Gaussian Avatar通常能更清晰地保留特写区域的外观细节，包括头部/面部（天蓝色方框）和服装（橙色方框）。所有方法均采用实时立体渲染技术，支持虚拟现实环境中的交互式虚拟形象操控。

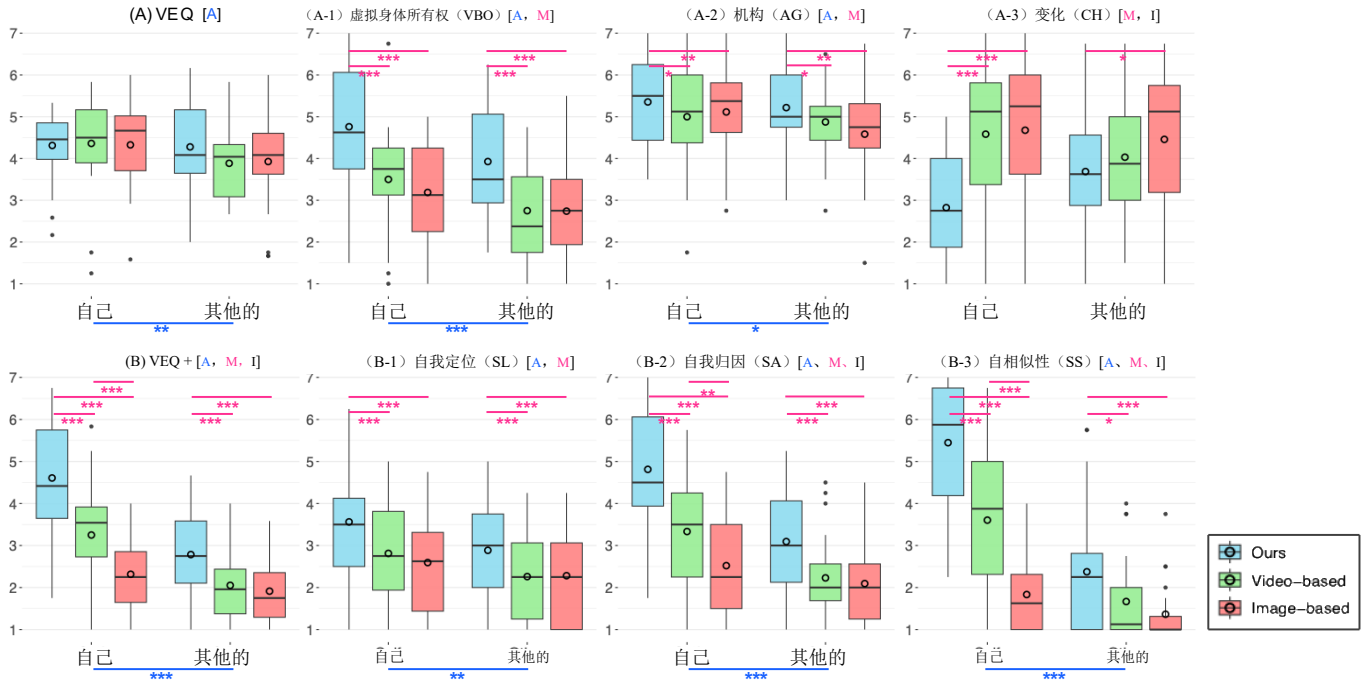


图6：(A)–(A-3)虚拟身体体验问卷及其子量表；(B)–(B-3) VEQ+及其子量表的结果。（A和M：分别表示化身类型与呈现方式的显著主效应；I：化身类型与呈现方式的显著交互效应）

视频与图像之间存在显著差异 ($p < .001$)，而视频与图像之间的差异则不显著。

SL显示出虚拟形象类型 ($F_{1, 23} = 9.389, p = .005, \eta^2_{p} = .290$) 与方法 ($F_{2, 46} = 12.948, p < .001, \eta^2_{p} = .360$) 的显著主效应。事后比较表明：自我虚拟形象的评分高于其他虚拟形象，且Ours虚拟形象的SL值显著更高。

而视频 ($p = .001$) 与图像 ($p < .001$) 之间存在显著差异，但视频与图像无显著差异。未发现交互效应。SA分析显示虚拟形象类型 ($F_{1, 23} = 45.904, p < .001, \eta^2_{p} = .666$) 和方法 ($F_{2, 46} = 44.525, p < .001, \eta^2_{p} = .659$) 均具有显著主效应，同时交互效应亦显著 ($F_{2, 46} = 10.653, p < .001, \eta^2_{p} = .317$)。事后分析表明自我类型的效果高于他人类型；我方法

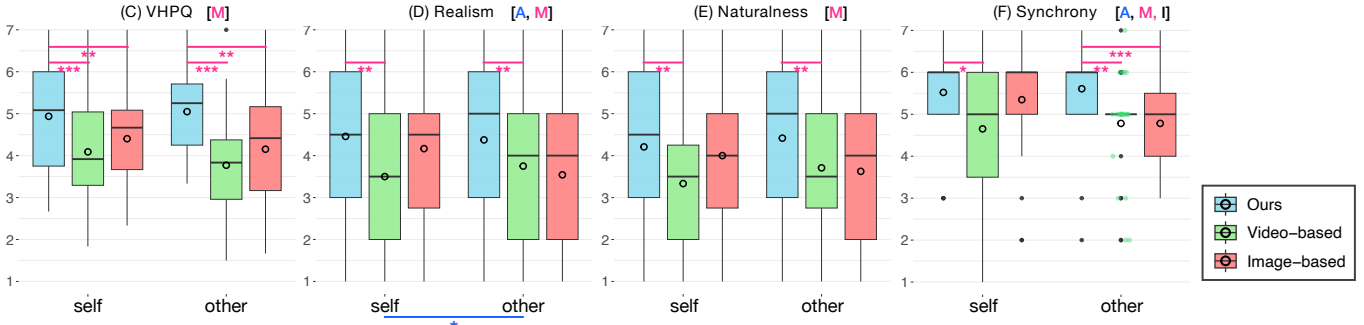


Fig. 7: Results for (C) Virtual Human Plausibility Questionnaire; (D)–(F) Additional subjective ratings. In (F), green jittered points indicate individual responses for the video-based method in the other avatar type, shown to complement the box plot where the interquartile range collapsed to a single value. (A and M: significant main effect of avatar type and method, respectively; I: significant interaction effect between avatar type and method)

was rated significantly higher than *video* and *image* (all $p < .001$), and *video* was also higher than *image* ($p = .006$). In both the *self* and *other* conditions, *Ours* induced significantly higher scores than *video* and *image* (all $p \leq .001$), and *video* was higher than *image* but only in the *self* condition ($p = .005$). Similarly, analysis of SS revealed significant main effects of avatar type ($F_{1,23} = 71.296$, $p < .001$, $\eta_p^2 = .756$), method ($F_{2,46} = 63.800$, $p < .001$, $\eta_p^2 = .735$), and their interaction ($F_{2,46} = 25.390$, $p < .001$, $\eta_p^2 = .525$). Post-hoc comparisons first indicated that *self* type was higher than *other* type. Moreover, *Ours* had significantly higher SS than *video* and *image*, and *video* was also higher than *image* (all $p < .001$). Within the *self* type, *Ours* perceived more similar with the participants than both *video* and *image*, and *video* was rated higher than *image* (all $p < .001$). In the *other* type, *Ours* also scored higher similarity than *video* ($p = .027$) and *image* ($p < .001$), while *video* and *image* had no difference.

5.1.2 Virtual Avatar Plausibility and Motion Behavior

VHPQ. The results regarding the VHPQ and self-reported measures are in Figure 7. With respect to the ABP in VHPQ, it showed a reliable Cronbach’s alpha ($\alpha_{VHPQ} = .900$). A main effect of method was observed ($F_{2,46} = 13.126$, $p < .001$, $\eta_p^2 = .363$), with *Ours* induced higher appearance and behavior plausibility than *video* ($p = .001$) and *image* ($p = .005$). No effects of avatar type or interaction were found.

Self-reported Motion Behavior. For motion realism, significant main effects of avatar type ($F_{1,23} = 4.656$, $p = .042$, $\eta_p^2 = .168$) and method ($F_{2,46} = 5.244$, $p = .009$, $\eta_p^2 = .186$) were observed, while the interaction effect was not significant. The pairwise post-hoc tests indicated that *self* avatars were higher than *other* avatars, and *Ours* induced higher behavior realism than *video* ($p = .009$). No other differences were significant. Regarding naturalness, a significant main effect of method was only found ($F_{2,46} = 4.950$, $p = .011$, $\eta_p^2 = .177$). Similar with the realism, post-hoc analysis revealed that *Ours* had significantly higher naturalness than *video* ($p = .009$), while *image* did not differ from either. No effect of avatar type or interaction was found. Lastly, one participant’s data contained an error and was excluded from the synchrony analysis. Synchrony showed a main effect of method ($F_{2,44} = 10.604$, $p < .001$, $\eta_p^2 = .325$), and an interaction with avatar type ($F_{2,44} = 3.459$, $p = .040$, $\eta_p^2 = .136$). In post-hoc analysis, it was revealed that *Ours* was rated as having higher motion synchrony than both *video* and *image* (all $p < .001$), and also *video* was higher than *image* ($p = .012$). In the *self* condition, *Ours* method had higher values than *video* ($p = .031$). In the *other* condition, *Ours* was higher than both *video* ($p = .002$) and *image* ($p < .001$), while other pairs were not different.

5.2 Quantitative Results

Table 1 summarizes the quantitative evaluation across different conditions. Our method (VRGA) achieves the best performance in all

Table 1: Quantitative comparison of rendering quality across different conditions. Video-based avatars are constructed following [47, 48, 62], and image-based avatars are constructed following [3, 28].

Condition	L1 ↓	LPiPS ↓	PSNR ↑	SSIM ↑
Video-based Avatar	0.042	0.099	16.33	0.927
Image-based Avatar	0.043	0.103	16.74	0.917
VRGA (Ours)	0.033	0.082	17.92	0.934

Table 2: Rendering performance across different resolutions with and without Binocular Batching (BB). Binocular Batching enables efficient real-time stereoscopic rendering. Resolution indicates the per-eye resolution, and FPS is measured under binocular rendering.

Resolution	Binocular Batching	LPiPS ↓	Render Time	FPS ↑
512×512	w/ BB	0.085	21.78 ms	45.91
	w/o BB		37.47 ms	26.69
768×768	w/ BB	0.084	23.19 ms	43.12
	w/o BB		38.33 ms	26.09
1024×1024	w/ BB	0.082	25.45 ms	39.29
	w/o BB		38.80 ms	25.77

metrics, demonstrating consistent advantages over both baselines. Between the two baselines, the video-based avatar shows better results than the image-based avatar in terms of L1, LPiPS, and SSIM.

Table 2 presents image similarity, rendering time, and FPS across different resolutions with and without Binocular Batching (BB). By exploiting the fact that stereoscopic rendering in VR shares SMPL-X parameters and dynamic 3DGS attributes, our system renders both camera views simultaneously with a batch size of two, enabling high-quality avatar rendering in real time. For the user study, we employed Binocular Batching at a resolution of 1024×1024, which provided real-time performance and visual quality.

Figure 5 provides representative qualitative examples that visually corroborate these quantitative improvements, showing clearer appearance details for VRGaussianAvatar under identical HMD-only control. More qualitative results are provided in Appendix B.

6 DISCUSSION

The goal of this work is to enable practical, high-fidelity avatar representation for VR applications by addressing the limitations of existing video- and image-based approaches. To this end, we introduce VRGaussianAvatar, an HMD-only controllable avatar pipeline that integrates 3D Gaussian Splatting into an interactive VR system, and evaluate whether

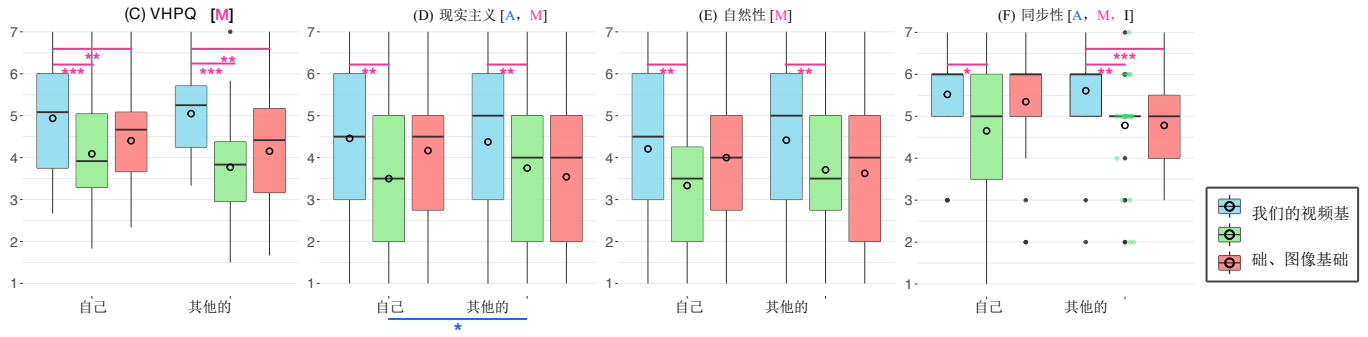


图7: (C) 虚拟人类可信度问卷结果; (D)–(F) 额外主观评分。在(F)图中, 绿色抖动点表示基于视频方法在另一种虚拟形象类型中的个体响应, 该图用于补充箱线图(箱线图中四分位距收缩至单一数值)。(A和M: 分别表示虚拟形象类型与方法的显著主效应; I: 虚拟形象类型与方法之间的显著交互效应)

其评分显著高于视频和图像 (所有 $p < .001$), 且视频的评分也高于图像 ($p = .006$)。在自我和他人两种条件下, 我们的方案所获得的评分均显著高于视频和图像 (所有 $p \leq .001$); 而视频的评分仅在自我条件下高于图像 ($p = .005$)。类似地, SS分析显示虚拟形象类型 ($F_{1, 23} = 71.296, p < .001, \eta^2_{2p} = .756$)、方法 ($F_{2, 46} = 63.800, p < .001, \eta^2_{2p} = .735$) 及其交互作用 ($F_{2, 46} = 25.390, p < .001, \eta^2_{2p} = .525$) 均具有显著主效应。事后比较首先表明自我类型的评分高于他人类型。此外, 我们的感知相似度显著高于视频和图像, 而视频的相似度也高于图像 (所有 $p < .001$)。在自我类型中, 我们的被参与者感知的相似度高于视频和图像, 且视频的评分也高于图像 (所有 $p < .001$)。在其他类型中, 我们的相似度评分也高于视频 ($p = .027$) 和图像 ($p < .001$), 而视频与图像之间无显著差异。

5.1.2 虚拟化身的可信度与运动行为

VHPQ.关于 VHPQ 和自我报告测量结果的数据见图7。就 VHPQ 中的 ABP而言, 其Cronbach's α 系数具有可靠性 ($\alpha_{VHPQ} = .900$)。观察到方法的主效应 ($F_{2, 46} = 13.126, p < .001, \eta^2_{2p} = .363$), 其中我们使用的方法比视频 ($p = .001$) 和图像 ($p = .005$) 能产生更高的场景呈现可信度和行为合理性。未发现虚拟角色类型或交互方式的影响。

自报运动行为。在运动真实感方面, 观察到虚拟形象类型 ($F_{1, 23} = 4.656, p = .042, \eta^2_{2p} = .168$) 与方法 ($F_{2, 46} = 5.244, p = .009, \eta^2_{2p} = .186$) 的主效应均显著, 而交互效应不显著。成对事后检验表明: 自我虚拟形象的表现优于其他虚拟形象; 我们的虚拟形象比视频虚拟形象表现出更高的行为真实感 ($p = .009$), 其他差异均不显著。在自然度方面, 仅发现方法的主效应显著 ($F_{2, 46} = 4.950, p = .011, \eta^2_{2p} = .177$)。与真实感类似, 事后分析显示: 我们的虚拟形象自然度显著高于视频虚拟形象 ($p = .009$), 而图像虚拟形象与两者均无显著差异。未发现虚拟形象类型或交互效应的影响。最后, 一名参与者的数据存在错误, 因此被排除在同步性分析之外。同步性分析显示方法存在主效应 ($F_{2, 44} = 10.604, p < .001, \eta^2_{2p} = .325$), 并与虚拟形象类型存在交互作用 ($F_{2, 44} = 3.459, p = .040, \eta^2_{2p} = .136$)。事后分析表明: Ours的运动同步性评分高于视频和图像 (所有 $p < .001$); 同时视频的同步性评分也高于图像 ($p = .012$)。在自我条件下, Ours方法的评分高于视频 ($p = .031$); 在其他条件下, Ours的评分高于视频 ($p = .002$) 和图像 ($p < .001$), 而其余配对组间无显著差异。

5.2 定量结果

表1总结了不同条件下的定量评估结果。我们的方法 (VRGA) 在所有测试中均取得了最佳性能。

表1: 不同条件下渲染质量的定量比较。基于视频的虚拟形象根据[47、48、62]构建, 而基于图像的虚拟形象则根据[3、28]构建。

条件	L1 ↓	LPiPS ↓	PSNR ↑	SSIM ↑
基于视频的虚拟形象	0.042	0.099	16.33	0.927
基于图像的虚拟形象	0.043	0.103	16.74	0.917
VRGA (我们的)	0.033	0.082	17.92	0.934

表2: 采用与不采用双眼批量处理 (BB) 时不同分辨率下的渲染性能。双眼批量处理可实现高效的实时立体渲染。分辨率指每只眼睛的分辨率, 帧率 (FPS) 是在双眼渲染模式下测量的。

分辨率	双筒望远镜 批量处理	LPiPS ↓	渲染 时间	FPS ↑
512×512	w/ BB	0.085	21.78 毫秒	45.91
	w/o BB		37.47 毫秒	26.69
768×768	w/ BB	0.084	23.19 毫秒	43.12
	w/o BB		38.33 毫秒	26.09
1024×1024	w/ BB	0.082	25.45 毫秒	39.29
	w/o BB		38.80 毫秒	25.77

各项指标均显示该方案相较于两种基准方法均具有持续优势。在L1、LPiPS和SSIM三项指标上, 基于视频的虚拟形象表现均优于基于图像的虚拟形象。

表2展示了采用与不采用双眼批量渲染 (BB) 技术时, 在不同分辨率下的图像相似度、渲染时间和帧率 (FPS)。通过利用虚拟现实立体渲染共享 SMPL-X参数和动态3DGS属性这一特性, 我们的系统以每批两个视角的规模同时渲染两个摄像头视图, 从而实现实时高质量虚拟形象渲染。在用户研究中, 我们采用1024×1024分辨率下的双眼批量渲染技术, 该方案既保证了实时性能又实现了优异的视觉质量。

图5展示了具有代表性的定性示例, 这些示例直观地验证了上述定量改进效果: 在仅使用头戴式显示器 (HMD) 的相同控制条件下, VR Gaussian Avatar的外观细节呈现得更为清晰。更多定性结果详见附录B。

6 讨论

本研究旨在通过克服现有基于视频和图像的方法的局限性, 为虚拟现实 (VR) 应用提供实用且高保真的虚拟形象表示方案。为此, 我们推出了VRGaussianAvatar——一种仅需头戴式显示器 (HMD) 即可控制的虚拟形象生成流程, 该流程将三维高斯贴片技术整合到交互式VR系统中, 并评估其可行性。

these technical contributions translate into perceptually meaningful improvements through a user study.

Our method produced stronger virtual embodiment than video and image-based approaches, particularly in body ownership and agency. Participants reported that the avatar looked and moved like their own body, with less alteration and greater resemblance. While aggregated VEQ scores did not differ significantly, subscales revealed that the sense of “my body” was more clearly enhanced with our method (“*felt the closest to how I actually move*,” P7). Further analyses of VEQ+ showed notable advantages in self-location and self-identification (self-attribution, and self-similarity). Participants described feeling that their body shifted into the avatar’s position in VR and that the avatar both resembled and responded to them. Even in other-representative conditions, where similarity scores were relatively low, our method still surpassed alternatives by producing more human-like and relatable avatars. By contrast, video-based avatars were described as “*overall texture seemed reflected but with unnatural movement*,” and image-based as “*only reflecting clothing color, not my actual features*” (P2).

Beyond embodiment, our system achieved the highest plausibility and realism, with more natural and synchronized movements than video-based avatars. Although some participants acknowledged that image-based avatars behaved smoothly or without latency (P13, P18), they emphasized that our method offered more consistent control and stronger human-likeness (“*moved naturally and felt alive*,” P15). Finally, these advantages translated into higher preference and practical usability. Most participants (79.2%) selected our method as the avatar they would most prefer and want to use in a virtual space, describing it as “*resembled me the most and felt natural*” (P20, P23). Even when evaluating “other” avatars, Ours was the most appealing to participants (50%) while maintaining higher plausibility and synchrony, underscoring its applicability for social VR scenarios. Taken together, these findings show that our method better represents virtual embodiment, identification, and realism than baseline approaches, providing a solid foundation for both self and other-representative avatars in VR.

These results indicate that the perceptual advantages observed in the user study may be associated with the technical design of the proposed pipeline. By separating real-time pose estimation from high-cost Gaussian rendering through the VR Frontend and GA Backend architecture, the system maintains stable and responsive control while preserving visual fidelity, which may help support users’ sense of embodiment and agency. In addition, Binocular Batching enables consistent stereoscopic rendering without redundant computation, allowing 3DGS-based avatars to remain interactive and perceptually plausible. As such, VRGaussianAvatar provides a practical foundation for deploying high-fidelity, controllable avatars in scenarios where both visual realism and embodied interaction play central roles; it is expected to show particularly high applicability in multi-user immersive environments in which users must continuously perceive both themselves and others as coherent agents (e.g., social VR, remote collaboration, or games). While further validation with larger samples and extended tasks is needed, the ability to support both self- and other-representative avatars within a single, efficient pipeline has the potential to broaden its applicability.

6.1 Limitations and Future Work

Our system has several limitations. First, because we do not integrate facial sensing signals, the avatar does not support dynamic facial expressions, gaze, or blinking. Second, our reconstruction assumes a controlled input image that is favorable for one-shot full-body reconstruction; performance may degrade under substantial occlusions or poses that deviate from the capture guideline. Third, while GPU-side rendering runs in real time on a desktop GPU, the end-to-end system latency is subject to streaming overhead. While our implementation runs on localhost, deploying the GA Backend over a remote network would increase end-to-end latency and may require bitrate–quality trade-offs (e.g., lower resolution or stronger compression) to maintain interactivity. Finally, our user study primarily relies on mirror-based observation and therefore does not directly evaluate embodiment outcomes from an egocentric (first-person) viewpoint.

Future work will aim to improve practicality and generalizability by addressing the limitations above. First, for social VR scenarios, we plan to support dynamic facial control by incorporating head-avatar reconstruction/control techniques (e.g., VOODOO XP [51]). Second, to reduce reliance on controlled capture, we will explore robustness strategies for more unconstrained inputs. Third, to move toward standalone deployment, we will investigate on-device avatar representation by adopting lightweight 3DGS avatar techniques (e.g., SqueezeMe [16]). Fourth, we will conduct additional user studies that directly evaluate embodiment from a first-person viewpoint.

6.2 Ethical Considerations

Because our system can generate high-fidelity avatars from a single image, it may be misused for non-consensual digital cloning or impersonation. To mitigate such risks, we restrict the integrated system to research and educational use and require written consent from the subject prior to avatar creation, including a clear specification of the intended use and duration. In addition, XR deployments that present reconstructed avatars should explicitly disclose the driving user to reduce deception risks. Our user study was conducted under Institutional Review Board (IRB) approval and followed standard consent and privacy procedures. If considering broader deployment, safeguards will likely require a combination of technical and procedural measures, such as consent protocols, identity verification, and standardized watermarking.

7 CONCLUSION

Building on these findings, we presented a system for real-time full-body 3DGS avatar representation in VR environments that can be controlled solely through internal sensors of commercial HMDs. Our framework integrates a VR Frontend, which estimates controllable full-body motion from these sensor signals, with a GA Backend that performs efficient stereoscopic rendering via Binocular Batching. This parallel design enables photorealistic and responsive avatar experiences without the need for additional external trackers, lowering the barrier for practical VR deployment. In addition, our quantitative evaluations and user studies confirm that VRGaussianAvatar not only achieves more photorealistic avatar representation compared to mesh-based avatar systems reconstructed from images and videos, but also provides higher levels of virtual embodiment and plausibility for users.

In summary, VRGaussianAvatar demonstrates that 3D Gaussian Splatting can be effectively extended to real-time, photorealistic, and HMD-only controllable avatar representation in VR. We believe this work highlights the feasibility of real-time 3DGS-based avatars and opens the door to richer and more engaging social interaction in immersive environments.

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这些技术贡献通过用户研究转化为具有感知意义的改进。

我们的方法比基于视频和图像的方法产生了更强的虚拟具身感，尤其是在身体所有权和自主性方面。参与者表示，虚拟形象的外观和动作都与他们自己的身体极为相似，且改动更少、逼真度更高。尽管综合 VEQ 评分未呈现显著差异，但子量表分析显示，我们的方法更显著增强了“我的身体”感知（“感觉最接近我实际的动作方式，P7”）。对 VEQ+ 的进一步分析表明，该方法在自我定位和自我认同（自我归因与自我相似性）方面具有显著优势。参与者描述称，在虚拟现实场景中他们的身体仿佛移动到了虚拟形象的位置，且虚拟形象既与他们相似又能对其做出反应。即使在其他相似度评分相对较低的场景中，我们的方法仍通过生成更人性化、更具代入感的虚拟形象超越了其他方案。相比之下，基于视频的虚拟形象被评价为“整体纹理较为真实，但动作显得不自然”，而基于图像的方法则被描述为“仅能反映服装颜色，无法呈现真实面部特征”（P2）。

除了虚拟具身感之外，我们的系统还实现了最高的可信度与真实感，其动作比基于视频的虚拟形象更加自然且同步。尽管部分参与者承认图像类虚拟形象运行流畅且无延迟（P13、P18），但他们强调我们的方法提供了更稳定的控制体验和更强的人类相似度（“动作自然，充满生命力” P15）。这些优势最终转化为更高的用户偏好与实际可用性：大多数参与者（79.2%）选择我们的方法作为虚拟空间中最理想的虚拟形象，并评价其“最贴近真实自我，操作体验最为自然”（P20、P23）。即使在评估“其他”类型虚拟形象时，我们的方案仍以50%的支持率最受青睐，同时保持更高的可信度与同步性，凸显了其在社交虚拟现实场景中的适用性。综合来看，本研究结果表明：相较于基线方法，我们的方法在虚拟具身感、身份认同度及真实感方面表现更优，为虚拟现实场景中既能反映自身特征又能体现他人形象的虚拟形象提供了坚实基础。

这些结果表明，用户研究中观察到的感知优势可能与所提出的流程技术设计密切相关。通过将实时姿态估计与高成本的高斯渲染分离（分别通过VR前端和GA后端架构实现），该系统在保持视觉保真度的同时，实现了稳定且响应迅速的控制，这有助于增强用户的具身感与自主性体验。此外，双目分批处理技术实现了无冗余计算的一致性立体渲染，使基于3DGS的虚拟形象保持交互性与视觉真实感。因此，VR高斯虚拟形象为在视觉真实性和具身交互均至关重要的场景中部署高保真、可控的虚拟形象提供了实用基础；预计其在需要用户持续将自身及他人视为连贯智能体的多用户沉浸式环境中（如社交虚拟现实、远程协作或游戏）将展现出极高的适用性。尽管仍需通过更大规模的样本和更复杂的任务进行进一步验证，但能够在单一高效的流程中同时支持自我代表性和他人代表性的虚拟化身，这一特性有望拓展其应用范围。

6.1 局限性与未来研究方向

我们的系统存在若干局限性。首先，由于未整合面部传感信号，虚拟形象无法实现动态面部表情、视线追踪或眨眼功能。其次，我们的重建算法基于可控输入图像，这种图像更有利于单次全身重建；在存在显著遮挡或姿势偏离采集规范的情况下，系统性能可能会下降。第三，虽然GPU端渲染可在桌面GPU上实时运行，但端到端系统的延迟会受到流式传输开销的影响。尽管我们的系统部署在本地主机上，但若通过远程网络部署GA后端，将增加端到端延迟，并可能需要在比特率与画质之间做出权衡（例如降低分辨率或加强压缩）以维持交互性。最后，我们的用户研究主要依赖于基于镜像的观察，因此未能从自我中心（第一人称）视角直接评估身体表征效果。

未来的工作将致力于通过解决上述局限性来提升系统的实用性和泛化能力。首先，在社交虚拟现实场景中，我们计划通过整合头部-化身重建/控制技术（例如voodoo XP[51]）来支持动态面部控制；其次，为降低对受控数据采集的依赖，我们将探索适用于更自由输入条件的鲁棒性策略；第三，为实现独立部署，我们将采用轻量级3DGS化身技术（例如SqueezeMe[16]）研究设备端化身表示方法；第四，我们将开展更多直接从第一人称视角评估具身体验的用户研究。

6.2 伦理考量

由于我们的系统能够仅凭单张图像生成高保真度的虚拟形象，因此可能被用于未经同意的数字克隆或冒名顶替行为。为降低此类风险，我们仅允许该集成系统用于科研和教育用途，并要求在创建虚拟形象前获得受试者的书面同意，其中需明确说明预期用途及使用时长。此外，展示重建虚拟形象的XR应用必须明确披露驱动用户信息，以降低欺骗风险。我们的用户研究已获得机构审查委员会（IRB）批准，并遵循标准的知情同意与隐私保护流程。若考虑更广泛的应用场景，安全防护措施可能需要结合技术手段与流程规范，例如制定明确的同意协议、实施身份验证机制以及采用标准化水印技术。

7 结论

基于这些研究成果，我们开发了一套适用于虚拟现实环境的实时全身3DGS虚拟形象呈现系统，该系统仅需通过商用头戴式显示器（HMD）的内置传感器即可实现操控。我们的框架整合了VR前端与遗传算法后端：前端根据传感器信号估算可控制的全身运动，后端则通过双眼批量渲染技术实现高效的立体视觉呈现。这种并行设计无需额外外部追踪器即可实现逼真且响应灵敏的虚拟形象体验，显著降低了实际部署虚拟现实系统的门槛。此外，定量评估与用户研究表明，VR Gaussian Avatar不仅比基于图像和视频重建的网格虚拟形象系统能呈现更逼真的虚拟形象，还能为用户带来更高层次的虚拟具身感与可信度。

总之，VR Gaussian Avatar 证明了三维高斯喷溅技术可以有效扩展至虚拟现实环境中实时、逼真且仅通过头戴式显示器（HMD）即可操控的虚拟形象呈现。我们认为，这项工作凸显了基于实时三维高斯喷溅技术的虚拟形象的可行性，并为沉浸式环境中更丰富、更具互动性的社交体验开辟了道路。

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VRGaussianAvatar: Integrating 3D Gaussian Splatting Avatars into VR

Appendix.

A IMPLEMENTATION DETAILS

A.1 VR System Configuration

The VR runtime consists of a Unity-based VR Frontend and a Python-based GA Backend running on the same local machine. The HMD is connected via a wired link to ensure stable high-bandwidth transmission. Key system parameters are summarized in Table 3.

Table 3: VR System and Rendering Hyperparameters.

Parameter	Value
<i>Rendering Backend (Python)</i>	
Render Resolution (per eye)	1024 × 1024
Field of View (FOV)	65.0°
Inter-pupillary Distance (IPD)	0.055 m
JPEG Compression Quality	80
Binocular Batching	Enabled (Single Pass)
<i>VR Frontend (Unity)</i>	
Target HMD	Meta Quest 3
Connection Type	Wired Link (USB 3.0)
Network Protocol	WebSocket (Localhost)
IK Solver	Final IK (VRIK)
Update Rate	Client-driven (Async)

A.2 Offline Reconstruction Pipeline in GA Backend

We utilize the Large Human Model (LHM-500M) [42] architecture for the single-image 3D Gaussian Splatting reconstruction. This architecture leverages powerful pre-trained encoders, specifically DINOv2 and Sapiens, to extract semantic and geometric features from a single RGB image, which are then decoded into 3D Gaussian parameters via a transformer-based network. For a more detailed description of the reconstruction pipeline and training strategy, please refer to LHM [42]. The detailed hyperparameters for the model architecture and training process are listed in Table 4.

Table 4: Hyperparameters for the Offline Reconstruction Pipeline.

Category	Parameter	Value
Model Architecture	Image Encoder	DINOv2-ViT-L/14 (Frozen)
	Fine Encoder	Sapiens-1B (Frozen)
	Transformer Config	5 Layers, 16 Heads, 1024 Hidden Dim
	Point Embeddings	1024 Dimensions
	Gaussian MLP	2 Layers, 512 Neurons, SiLU Activation
Training	Optimizer	AdamW ($\beta_1 = 0.9, \beta_2 = 0.95$)
	Learning Rate	4×10^{-5} (Cosine Schedule)
	Weight Decay	0.05
	Batch Size	4 (per GPU)
	Total Epochs	60
Loss Weights	Masked Pixel Loss	1.0
	Perceptual Loss	1.0
	Face ID Loss	0.05
	Offset Regularization	1000.0

A.3 Streaming Protocol and Latency

While our reference setup uses localhost, deploying the GA Backend over a remote network can introduce additional latency and bandwidth constraints. Maintaining interactive frame delivery in such settings may therefore require bitrate–quality trade-offs (e.g., stronger compression or lower resolution), depending on network conditions.

We implement a WebSocket-based streaming protocol in the reference setup. The VR Frontend sends per-frame HMD pose and SMPL-X parameters as JSON, and the GA Backend returns rendered stereo images as concatenated JPEG binaries. Table 5 reports the end-to-end latency breakdown measured under this localhost configuration.

Note that these stages run asynchronously in a pipelined manner. As a result, the achievable frame rate is primarily bounded by the GA Backend rendering time (and encoding), while the end-to-end latency reflects the full pipeline delay from tracking to display. We report FPS in Table 2 using the GA Backend throughput (rendering + encoding).

B MORE QUALITATIVE RESULTS AND CHALLENGING CASES

This section provides additional qualitative comparisons of VRGaussianAvatar against baseline mesh avatar representations under identical HMD-only control. We highlight representative regions where our 3DGS avatar preserves fine appearance cues (e.g., facial details and clothing texture), and we include a single challenging case focused on hands. Representative examples are shown in Figure 8.

Hands are particularly difficult to render sharply in our current pipeline. Compared to other body regions, hands have a higher kinematic freedom and undergo larger pose changes during natural interaction. However, the underlying 3DGS hand representation is reconstructed with a finite Gaussian density, which can be insufficient to stably represent fast, highly articulated deformations. As a result, our method can exhibit localized blurring around fingers during motion, and this artifact can be more noticeable than in relatively rigid regions or in mesh-based reconstructions.

As future work, we plan to incorporate a dynamic hand pose representation and integrate a VR-oriented 3DGS hand model that is designed for frequent, highly articulated hand motions.

VR高斯虚拟形象：将三维高斯点阵虚拟形象集成到虚拟现实环境中

附录。

A 实施详情

A.1 VR系统配置

该VR运行环境由基于Unity的VR前端和基于Python的遗传算法（GA）后端组成，两者运行在同一台本地计算机上。头戴式显示器（HMD）通过有线连接以确保稳定的高带宽传输。关键系统参数汇总于表3。

表3：VR系统及渲染超参数。

参数	价值
渲染后端 (Python)	
渲染分辨率（每只眼睛）视场（FOV）	1024 × 1024 65.0°
瞳孔间距（IPD）JPEG压缩质量双眼批量处理	0.055 m 80已启用（单次通过）
VR前端 (Unity)	
Target HMD 连接类型	Meta Quest 3 有线连接（USB 3.0）、WebSocket（本地主机）、最终IK（VRIK）、客户端驱动（异步）
网络协议 IK 解算器更新速率	

A.2 GA 后端的离线重建管道

我们采用大型人体模型（LHM -500M）[42]架构进行单图像三维高斯斑点重建。该架构利用强大的预训练编码器（特别是DINOv2和Sapiens），从单张RGB图像中提取语义和几何特征，随后通过基于Transformer的网络将这些特征解码为三维高斯参数。关于重建流程和训练策略的详细说明，请参阅 LHM [42]。模型架构及训练过程的详细超参数列于表4。

表4：离线重建流程的超参数。

类别	参数	价值
模型架构	图像编码器、精细编码器、Transformer 配置	DINOv2-ViT-L/14（冷冻）Sapiens-1B（冷冻） 5 层数；16；头部数量：10；隐藏维度：1024； 维度总数：1024
	点嵌入高斯 MLP	2 多层结构，512个神经元，SiLU激活函数
训练	学习率	AdamW ($\beta_1 = 0.9, \beta_2 = 0.95$) 4×10^{-5} （余弦调度）
	权重衰减 批量大小 总训练轮数	0.05 4（每块GPU） 60
损失权重	掩码像素损失、感知损失、Face ID 损失、偏移正则化	1.0 1.0 0.05 1000.0

A.3 流媒体协议与延迟

虽然我们的参考配置使用 localhost，但通过远程网络部署 GA 后端会引入额外的延迟和带宽限制。因此，在此类环境中保持交互式帧传输可能需要权衡比特率与画质（例如采用更强的压缩算法或更低的分辨率），具体取决于网络条件。

我们在参考配置中实现了基于 WebSocket 的流式传输协议。VR前端以 JSON 格式每帧发送头戴式显示器（HMD）的姿态数据和 SMPL-X 参数，而 GA 后端则返回拼接后的 JPEG 二进制文件形式的渲染立体图像。表 5 列出了在此 localhost 配置下测得的端到端延迟分布。

需要注意的是，这些阶段以异步流水线方式运行。因此，可实现的帧率主要受限于GA后端的渲染（及编码）时间，而端到端延迟则反映了从跟踪到显示的整个流水线延迟。我们在表2中使用GA后端吞吐量（渲染+编码）来报告FPS。

B 更多定性研究结果及具有挑战性的案例

本节在相同的仅头戴式显示器（HMD）控制条件下，对VR Gaussian Avatar与基线网格化身表示进行了进一步的定性比较。我们重点展示了3DGS化身能够保留精细外观特征（如面部细节和服装纹理）的代表性区域，并包含了一个针对手部的典型挑战案例。代表性示例如图8所示。

在当前的处理流程中，手部特征的清晰呈现尤为困难。与其他身体部位相比，手部具有更高的运动自由度，在自然交互过程中会发生更大幅度的姿态变化。然而，底层的3DGS手部模型采用有限的高斯密度进行重建，这种方法可能无法稳定地表征快速且高度关节化的形变。因此，我们的方法在手指运动时会出现局部模糊现象，这种伪影在相对刚性的区域或基于网格的重建结果中可能更为明显。

作为未来的研究方向，我们计划引入动态手部姿态表示，并整合一款面向虚拟现实（VR）的3DGS手部模型，该模型专为频繁且高度关节化的手部运动而设计。

Table 5: Latency breakdown of the streaming pipeline measured under the localhost configuration. Stages are grouped by where they execute (GA Backend vs. VR Frontend).

Stage	Location	Time (ms)
Tracking & inverse kinematics	VR Frontend	~ 1
Localhost communication (request/response)	GA Backend $\leftrightarrow$ VR Frontend	< 1
Stereo rendering (Binocular Batching)	GA Backend	$\sim 23\text{-}27$
Image encoding (JPEG, multi-threaded)	GA Backend	$\sim 1\text{-}2$
Image decoding & display	VR Frontend	$\sim 5\text{-}10$
End-to-end latency	–	$\sim 30\text{-}41$

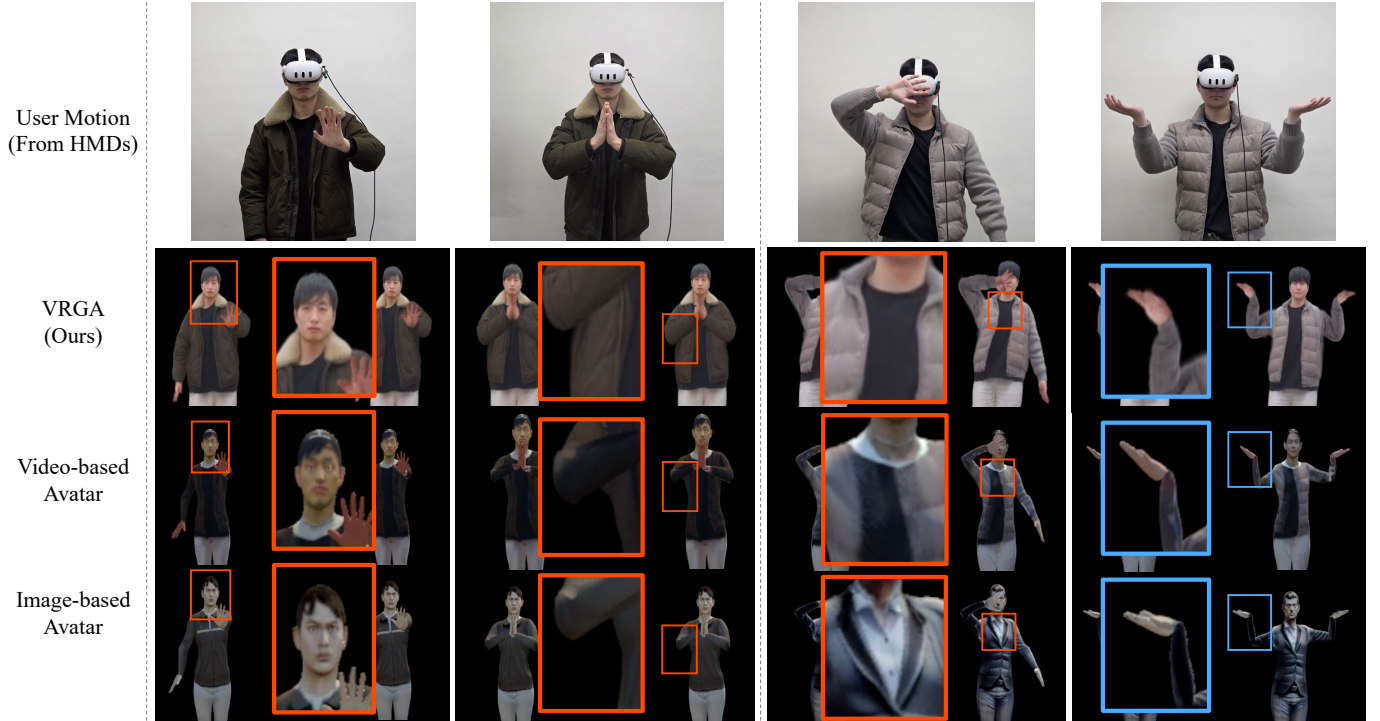


Fig. 8: **More qualitative results.** Orange boxes indicate representative close-ups where our method preserves facial and clothing details. Sky-blue boxes highlight a challenging hand region, discussed in [Appendix B](#).

表5：在localhost配置下测得的流式处理管道延迟分布。各处理阶段按执行位置分组（GA后端 vs. VR前端）。

阶段	位置	时间（毫秒）
跟踪与逆运动学	VR Frontend	~ 1
本地主机通信（请求/响应）	GA 后端 # VR 前端	< 1
立体渲染（双眼批量处理）	GA 后端	~ 23-27
图像编码（JPEG，多线程）	GA 后端	~ 1-2
图像解码与显示	VR Frontend	~ 5-10
端到端延迟	—	~ 30-41

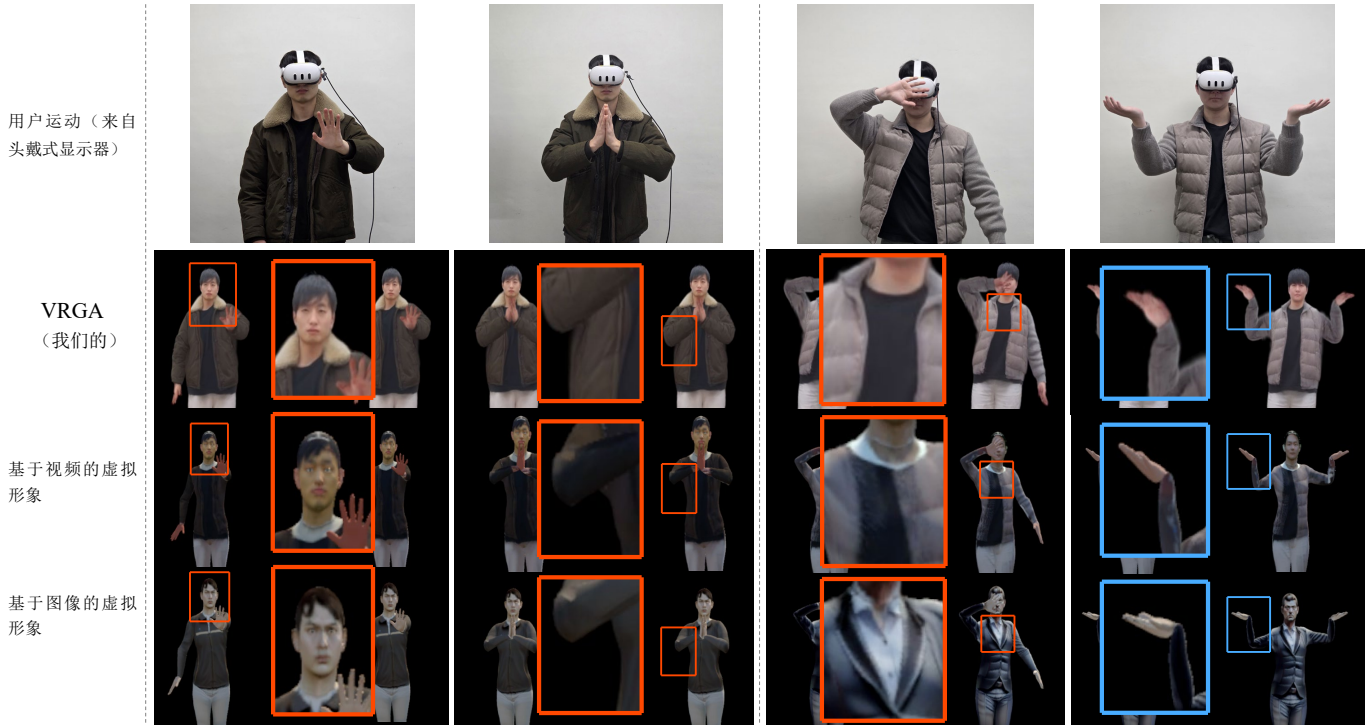


图8：更具定性的结果。橙色方框标示了具有代表性的特写图像，其中我们的方法保留了面部和衣物细节；天蓝色方框突出了一个具有挑战性的手部区域，详见附录B。